Imperial College London

05 December 2024

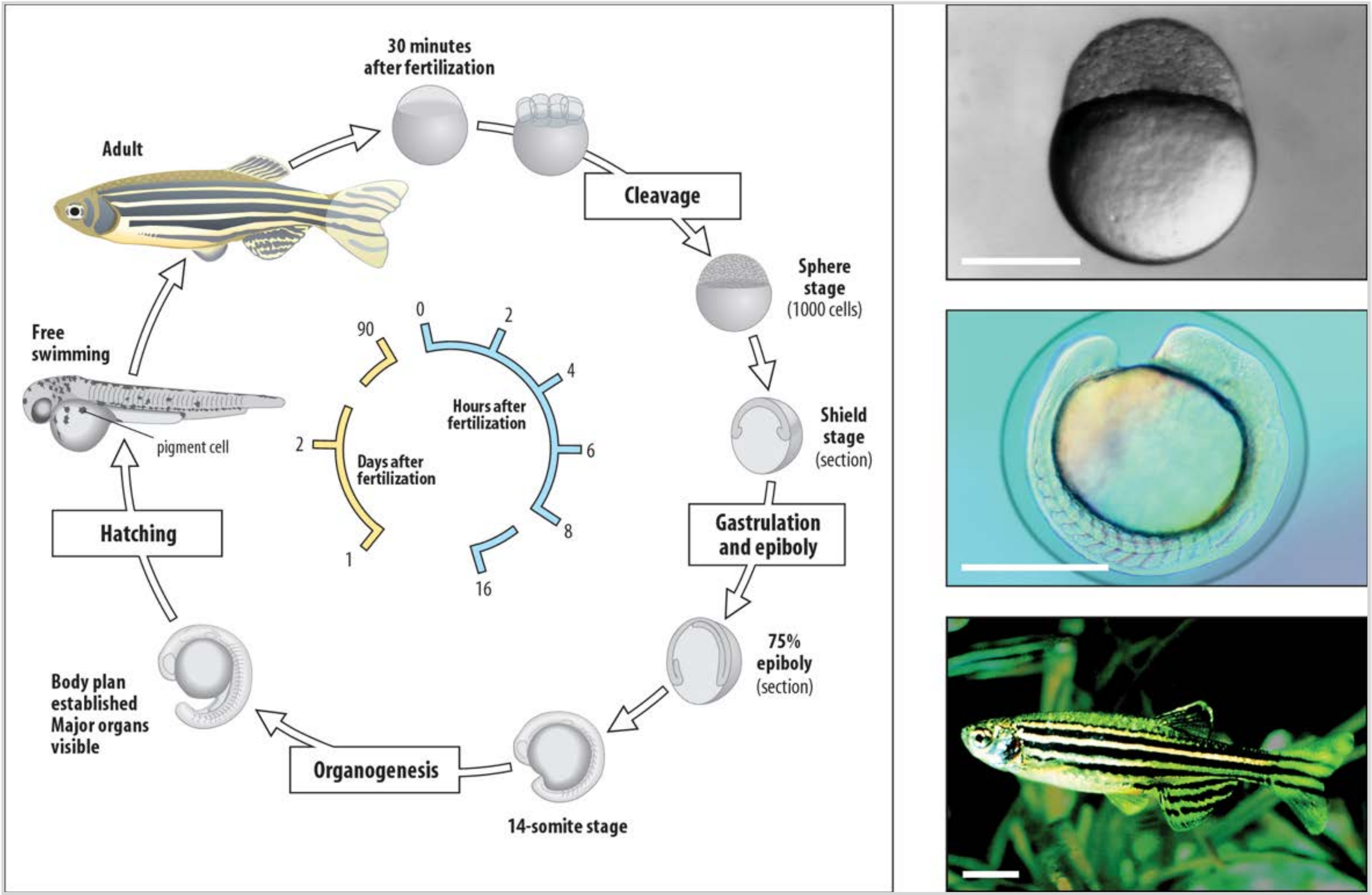
Advanced Imaging Technologies in Systems Biology and Biomedical Research

Periklis (Laki) Pantazis

What we covered

- Quantitative imaging of morphogen dynamics in zebrafish (Danio rerio)
- Reaction-diffusion (Turing) model of pattern generation

Zebrafish development

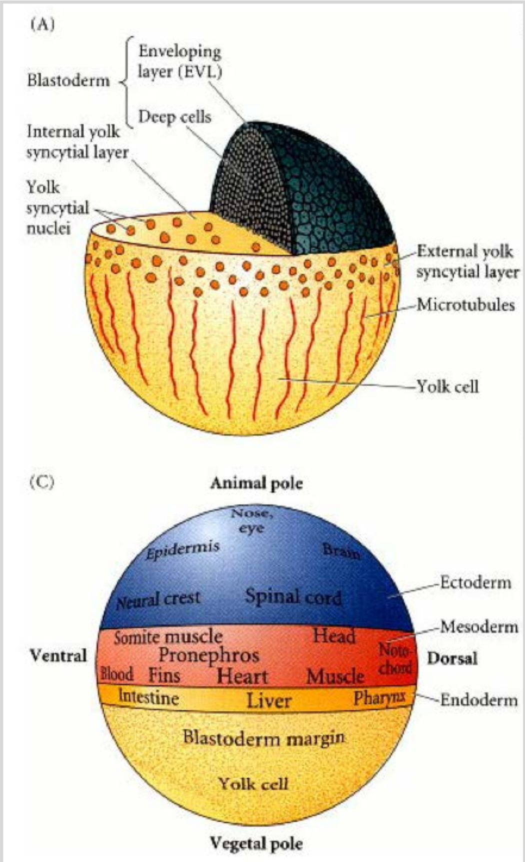


Zebrafish development: all together

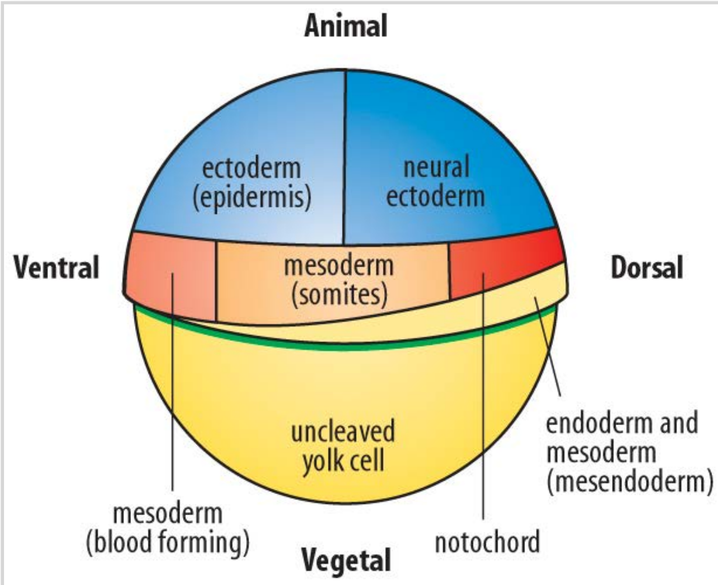
Danio Rerio
(Zebrafish)

Zebrafish development: fate maps

prior to gastrulation



at early gastrula stage



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LETTERS

Fgf8 morphogen gradient forms by a source-sink mechanism with freely diffusing molecules

Shuizi Rachel Yu[1,3]*, Markus Burkhardt[2,3]*, Matthias Nowak[1,3], Jonas Ries[2,3], Zdeněk Petrášek[2,3], Steffen Scholpp[1]†, Petra Schwille[2,3] & Michael Brand[1,3]

It is widely accepted that tissue differentiation and morphogenesis in multicellular organisms are regulated by tightly controlled concentration gradients of morphogens[1,2]. How exactly these gradients are formed, however, remains unclear[3–12]. Here we show that Fgf8 morphogen gradients in living zebrafish embryos are established and maintained by two essential factors: fast, free diffusion of single molecules away from the source through extracellular space, and a sink function of the receiving cells, regulated by receptor-mediated endocytosis. Evidence is provided by directly examining single molecules of Fgf8 in living tissue by fluorescence correlation spectroscopy, quantifying their local mobility and concentration with high precision. By changing the degree of uptake of Fgf8 into its target cells, we are able to alter the shape of the Fgf8 gradient. Our results demonstrate that a freely diffusing morphogen can set up concentration gradients in a complex multicellular tissue by a simple source-sink mechanism.

Morphogens are signalling molecules that can induce distinct cellular responses in a concentration-dependent manner[1,2]. Many examples have previously been described, including fibroblast growth factor 8 (Fgf8) in zebrafish embryonic development[2,13–18]. How morphogens such as Fgf8 move through a tissue and set up concentration gradients remains debated. In the 1970s, Francis Crick proposed that simple diffusion and spatially uniform degradation is sufficient for setting up gradients[3]. This mechanism has been considered in various systems[6–11] and it has been shown that a random walk model can accurately describe the DPP morphogen gradient in *Drosophila* wing disks[10]. Moreover, studies have suggested that morphogens can be sequestered from target tissue by receptor-mediated removal[12,18]. However, a full biophysical description of morphogen gradient formation is still lacking. Meanwhile, alternative models including receptor-aided bucket-brigade mechanism[4] and directed transport[11] have been proposed for morphogen spreading. Here we report a fluorescence correlation spectroscopy (FCS)-based assay for examining single-molecule dynamics of Fgf8 in living zebrafish embryos (Fig. 1a–c and Supplementary Information)[19,20]. By directly measuring its mobility and concentration gradient, we determine the mechanism of Fgf8 morphogen gradient formation (Supplementary Fig. 1).

We first tagged Fgf8 with enhanced green fluorescent protein (Fgf8–EGFP; Fig. 2a). At equimolar concentrations, the fusion protein showed a similar capacity to induce target gene expression as untagged Fgf8 (Supplementary Fig. 2a, b). Fgf8–EGFP was readily taken up by target cells, and was recruited onto cell membranes by Fgf receptors (Fig. 2a–e)[21]. In contrast, a negative control of secreted EGFP (secEGFP) was not. Furthermore, we verified by western blot that the full-length fusion protein was the only EGFP-containing species in the extracellular space of *fgf8-egfp* messenger-RNA-injected embryos (Supplementary Fig. 2c).

To investigate how Fgf8 spreads into its target tissue, we took FCS measurements in the extracellular space of gastrulating zebrafish embryos that had a restricted source of the fluorescent protein (Figs 1 and 2b, c and Methods Summary). The FCS autocorrelation curves fit well to a three-dimensional diffusion model, but not with alternative models such as directional active transport only (Fig. 3a and Supplementary Information). Most of the molecules (91%) had a diffusion coefficient ($53\,\mu m^2\,s^{-1}$) that was on the same order of magnitude as that of secEGFP ($86\,\mu m^2\,s^{-1}$; Fig. 3a, b and Supplementary Fig. 3a, b). Fgf8–EGFP moved slightly slower than secEGFP, as is expected for a larger molecule of about double the molecular mass. For a receptor-mediated bucket-brigade mechanism[4], we would expect most of the ligand to have a diffusion coefficient similar to that of the lateral mobility of the Fgf receptor in the cell membrane, which is about 100-fold smaller[22] than the one we measured for Fgf8, thus arguing against this mechanism for Fgf8 spreading. The Fgf8–EGFP diffusion coefficient in the zebrafish embryos is two orders of magnitude larger than that reported for DPP ($0.10\,\mu m^2\,s^{-1}$) in *Drosophila* wing discs[10]. This supports the notion that the two morphogens have very different propagation mechanisms in their respective target tissues. DPP is transported by transcytosis involving repeated cycles of endocytosis and exocytosis through target cells[5,10]. This is likely to occur much slower than the free diffusion of Fgf8–EGFP in the extracellular space.

In addition, we analysed recombinant Fgf8 protein labelled with the organic dye Cy5. The autocorrelation curve of Fgf8–Cy5 fitted

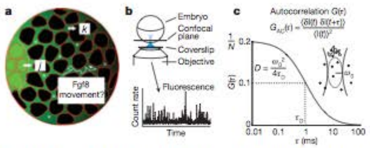

Figure 1 | FCS setup to investigate morphogen characteristics. a, Schematic section of a gastrulating zebrafish embryo. Morphogen is produced from a restricted source (green cells) with current *j*, spreads into target tissue (black cells) and is removed from the extracellular space with degradation rate *k*. Crosses indicate extracellular spaces where FCS measurements are taken. **b,** One-focus FCS setup. **c,** The autocorrelation (AC) function $G(\tau)$ has two key parameters: (1) the characteristic dwell time τ_D, which relates to the diffusion coefficient D; and (2) the amplitude $G(0)$, which is inversely proportional to the number of fluorescent particles N in the detection volume (Supplementary Information).

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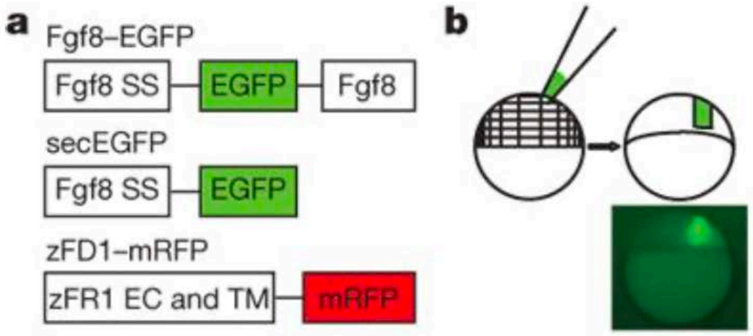
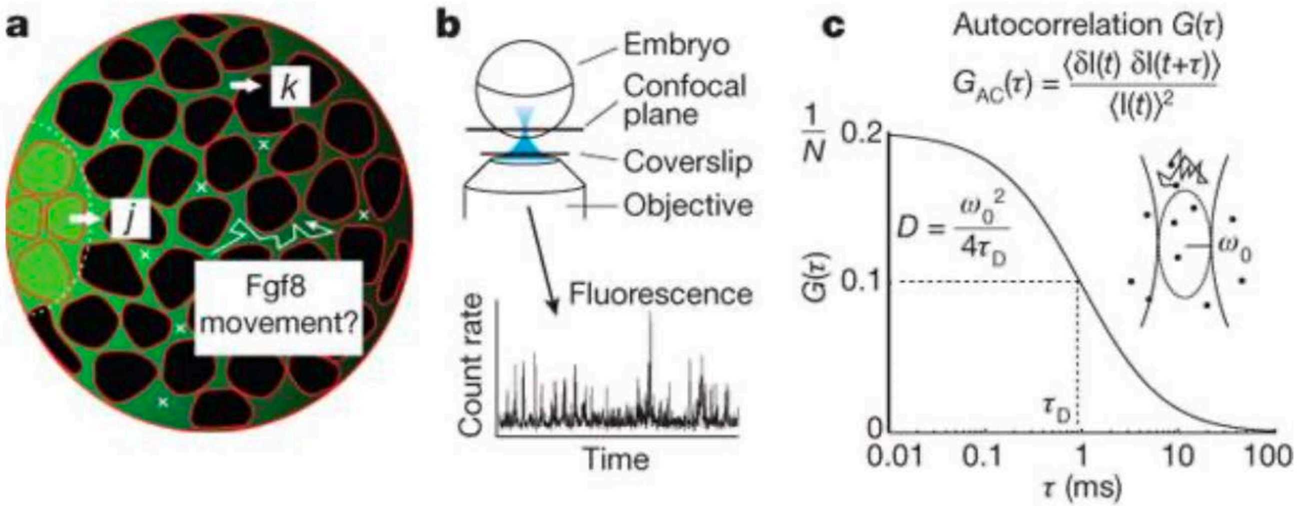
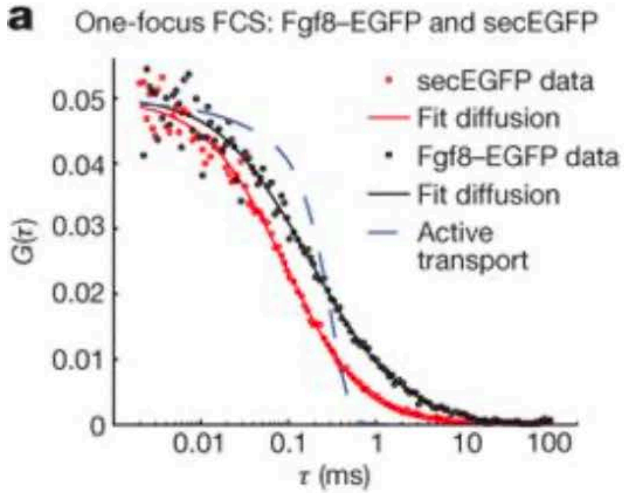
*These authors contributed equally to this work.

Experimental setup





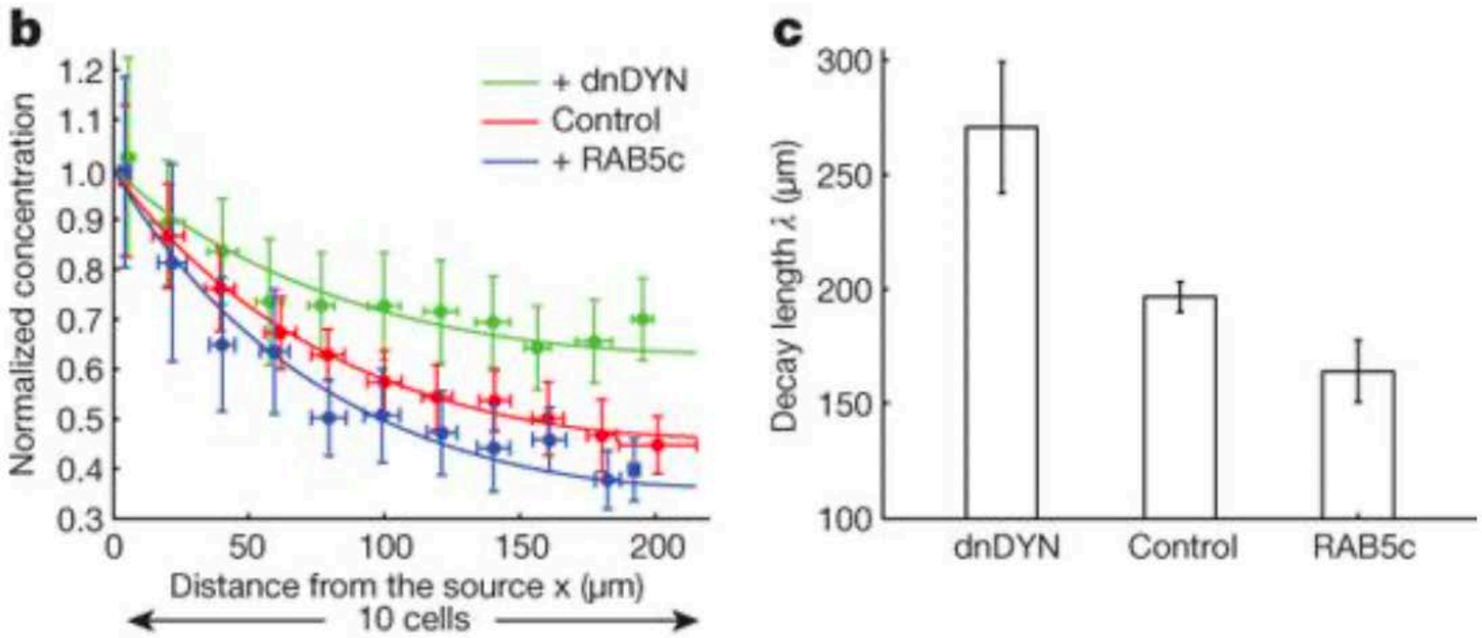
Free 3D Diffusion dominates the movement



b

	F (%)	Diffusion coefficient (μm^2 s^{-1}) Fast component	Slow component	MM (kDa)	n
secEGFP	99 ± 1	86 ± 11	NA	28	29
Fgf8–EGFP	91 ± 5	53 ± 8	4 ± 3	50	29
Cy5	99 ± 2	190 ± 6	NA	1	8
Fgf8–Cy5	80 ± 12	91 ± 26	7 ± 2	23	30

Modulating the Fgf8-gradient



Endocytosis regulates extracellular FGF8 concentration gradient
Data support a source sink model for gradient formation

Reaction-diffusion (Turing) model of pattern generation

THE CHEMICAL BASIS OF MORPHOGENESIS

By A. M. TURING, F.R.S. *University of Manchester*

(*Received* 9 *November* 1951—*Revised* 15 *March* 1952)

It is suggested that a system of chemical substances, called morphogens, reacting together and diffusing through a tissue, is adequate to account for the main phenomena of morphogenesis. Such a system, although it may originally be quite homogeneous, may later develop a pattern or structure due to an instability of the homogeneous equilibrium, which is triggered off by random disturbances. Such reaction-diffusion systems are considered in some detail in the case of an isolated ring of cells, a mathematically convenient, though biologically unusual system. The investigation is chiefly concerned with the onset of instability. It is found that there are six essentially different forms which this may take. In the most interesting form stationary waves appear on the ring. It is suggested that this might account, for instance, for the tentacle patterns on *Hydra* and for whorled leaves. A system of reactions and diffusion on a sphere is also considered. Such a system appears to account for gastrulation. Another reaction system in two dimensions gives rise to patterns reminiscent of dappling. It is also suggested that stationary waves in two dimensions could account for the phenomena of phyllotaxis.

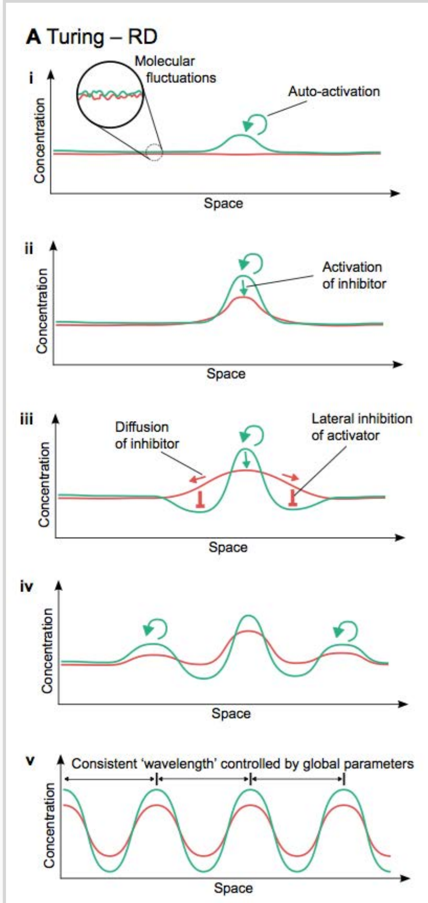
The purpose of this paper is to discuss a possible mechanism by which the genes of a zygote may determine the anatomical structure of the resulting organism. The theory does not make any new hypotheses; it merely suggests that certain well-known physical laws are sufficient to account for many of the facts. The full understanding of the paper requires a good knowledge of mathematics, some biology, and some elementary chemistry. Since readers cannot be expected to be experts in all of these subjects, a number of elementary facts are explained, which can be found in text-books, but whose omission would make the paper difficult reading.

1. A MODEL OF THE EMBRYO. MORPHOGENS

In this section a mathematical model of the growing embryo will be described. This model will be a simplification and an idealization, and consequently a falsification. It is to be hoped that the features retained for discussion are those of greatest importance in the present state of knowledge.

The model takes two slightly different forms. In one of them the cell theory is recognized but the cells are idealized into geometrical points. In the other the matter of the organism is imagined as continuously distributed. The cells are not, however, completely ignored, for various physical and physico-chemical characteristics of the matter as a whole are assumed to have values appropriate to the cellular matter.

With either of the models one proceeds as with a physical theory and defines an entity called 'the state of the system'. One then describes how that state is to be determined from the state at a moment very shortly before. With either model the description of the state consists of two parts, the mechanical and the chemical. The mechanical part of the state describes the positions, masses, velocities and elastic properties of the cells, and the forces between them. In the continuous form of the theory essentially the same information is given in the form of the stress, velocity, density and elasticity of the matter. The chemical part of the state is given (in the cell form of theory) as the chemical composition of each



undetectable in chondrocytes; CBFβ is expressed in all cell types at comparable levels (figs. S13 and S14). Fluorescent immunostaining of mouse joint tissue sections also showed coexpression of CBFβ and RUNX1 in cartilage cells (fig. S15). These results are consistent with previous findings about the roles of the genes in chondrogenesis and osteogenesis, and lead us to speculate that RUNX1, rather than RUNX2 or RUNX3, is probably the downstream effector of KGN. When we knocked down RUNX1 expression in hMSCs with shRNAs, KGN-induced chondrocyte differentiation was blocked (fig. S16). Furthermore, *RUNX2* transcription has been shown to be suppressed by RUNX2 through an autoregulation mechanism in different types of cells (*31*). Given the fact that all RUNX family members recognize the same DNA sequence, it is plausible that the RUNX1-CBFβ complex also suppresses *RUNX2* transcription, thus keeping RUNX2 at a relatively low level, which inhibits osteoblast and terminal chondrocyte differentiation.

The development of disease-modifying osteoarthritis drugs has to date focused on (i) slowing the erosion of cartilage by inhibiting matrix-degrading enzymes; (ii) protecting chondrocytes from catabolic activity with antagonists of interleukin-1β and TNF-α or inhibitors of p38, MEK, and caspases; and (iii) the development of anabolic agents, such as FGF18, which promote chondrocyte and chondrocyte progenitor cell proliferation (*3*, *32*, *33*). This work is based on small-molecule effectors of endogenous stem cell populations and may prove synergistic with other efforts currently in development. KGN functions by binding FLNA and disrupting its interaction with CBFβ, which in turn modulates the RUNX family of transcription factors. Recently, two other examples have been reported (*34*, *35*) in which small druglike molecules selectively regulate transcription factor subcellular localization and downstream transcriptional activities. Thus, small molecules appear to be useful tools for selectively modulating gene transcription in cells and whole organisms, and in this case have provided new insights into chondrocyte biology that may lead to novel therapies for the treatment of this degenerative disease.

References and Notes

1. J. Y. Reginster, N. G. Khaltaev, *Rheumatology (Oxford)* **41** (suppl. 1), 1 (2002).
2. M. B. Goldring, S. R. Goldring, *Ann. N.Y. Acad. Sci.* **1192**, 230 (2010).
3. D. J. Hunter, *Nat. Rev. Rheumatol.* **7**, 13 (2011).
4. P. Charbord, *Hum. Gene Ther.* **21**, 1045 (2010).
5. M. F. Pittenger *et al.*, *Science* **284**, 143 (1999).
6. S. P. Grogan, S. Miyaki, H. Asahara, D. D. D'Lima, M. K. Lotz, *Arthritis Res. Ther.* **11**, R85 (2009).
7. S. Koelling *et al.*, *Cell Stem Cell* **4**, 324 (2009).
8. G. N. Smith Jr., *Front. Biosci.* **11**, 3081 (2006).
9. M. Rutgers, D. B. Saris, W. J. Dhert, L. B. Creemers, *Arthritis Res. Ther.* **12**, R114 (2010).
10. S. J. Read, A. Dray, *Expert Opin. Investig. Drugs* **17**, 619 (2008).
11. P. M. van der Kraan, E. L. Vitters, L. B. van de Putte, W. B. van den Berg, *Am. J. Pathol.* **135**, 1001 (1989).
12. S. S. Glasson, T. J. Blanchet, E. A. Morris, *Osteoarthritis Cartilage* **15**, 1061 (2007).
13. K. P. Pritzker *et al.*, *Osteoarthritis Cartilage* **14**, 13 (2006).
14. L. S. Lohmander, T. Saxne, D. K. Heinegård, *Ann. Rheum. Dis.* **53**, 8 (1994).
15. M. Reijman *et al.*, *Arthritis Rheum.* **50**, 2471 (2004).
16. Y. Wang *et al.*, *Oncogene* **26**, 6061 (2007).
17. J. B. Gorlin *et al.*, *J. Cell Biol.* **111**, 1089 (1990).
18. J. T. Connelly, A. J. García, M. E. Levenston, *J. Cell. Physiol.* **217**, 145 (2008).
19. N. Yoshida *et al.*, *Mol. Cell. Biol.* **25**, 1003 (2005).
20. N. Adya, L. H. Castilla, P. P. Liu, *Semin. Cell Dev. Biol.* **11**, 361 (2000).
21. T. Komori, *J. Cell. Biochem.* **112**, 750 (2011).
22. D. Levanon, Y. Groner, *Oncogene* **23**, 4211 (2004).
23. Y. Wang *et al.*, *J. Bone Miner. Res.* **20**, 1624 (2005).
24. S. Wotton *et al.*, *Oncogene* **27**, 5856 (2008).
25. G. S. Stein *et al.*, *Oncogene* **23**, 4315 (2004).
26. T. Komori *et al.*, *Cell* **89**, 755 (1997).
27. F. Otto *et al.*, *Cell* **89**, 765 (1997).
28. S. Kamekura *et al.*, *Arthritis Rheum.* **54**, 2462 (2006).
29. D. Y. Soung *et al*, *J. Bone Miner. Res.* **22**, 1260 (2007).
30. P. M. van der Kraan, E. N. Blaney Davidson, W. B. van den Berg, *Arthritis Res. Ther.* **12**, 201 (2010).
31. H. Drissi *et al.*, *J. Cell. Biochem.* **86**, 403 (2002).
32. M. K. Lotz, V. B. Kraus, *Arthritis Res. Ther.* **12**, 211 (2010).
33. J. Clouet *et al.*, *Drug Discov. Today* **14**, 913 (2009).
34. S. Zhu *et al.*, *Cell Stem Cell* **4**, 416 (2009).
35. H. Wurdak *et al.*, *Proc. Natl. Acad. Sci. U.S.A.* **107**, 16542 (2010).

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Supplementary Materials
www.sciencemag.org/cgi/content/full/science.1215157/DC1
Materials and Methods
Figs. S1 to S16

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10.1126/science.1215157

Differential Diffusivity of Nodal and Lefty Underlies a Reaction-Diffusion Patterning System

Patrick Müller,[1*] Katherine W. Rogers,[1] Ben M. Jordan,[2†] Joon S. Lee,[1†] Drew Robson,[1] Sharad Ramanathan,[1,3] Alexander F. Schier[1,3*]

Biological systems involving short-range activators and long-range inhibitors can generate complex patterns. Reaction-diffusion models postulate that differences in signaling range are caused by differential diffusivity of inhibitor and activator. Other models suggest that differential clearance underlies different signaling ranges. To test these models, we measured the biophysical properties of the Nodal/Lefty activator/inhibitor system during zebrafish embryogenesis. Analysis of Nodal and Lefty gradients revealed that Nodals have a shorter range than Lefty proteins. Pulse-labeling analysis indicated that Nodals and Leftys have similar clearance kinetics, whereas fluorescence recovery assays revealed that Leftys have a higher effective diffusion coefficient than Nodals. These results indicate that differential diffusivity is the major determinant of the differences in Nodal/Lefty range and provide biophysical support for reaction-diffusion models of activator/inhibitor-mediated patterning.

In 1952, Alan Turing put forward the reaction-diffusion model, in which two interacting and diffusing species of molecules can generate complex patterns (*1*). Gierer and Meinhardt postulated that pattern formation in reaction-diffusion models requires a short-range activator that enhances both its own production and that of a long-range inhibitor (*2*) (Fig. 1A). Despite the prominence of reaction-diffusion models and the widespread occurrence of short-range activators and long-range inhibitors in development (*3*–*10*), it is unclear how differences in activator and inhibitor ranges arise in vivo. The classic reaction-diffusion models postulate that the inhibitor is more diffusive than the activator (text S1), but more recent studies suggest that differential signal clearance might be a major determinant of differences in signaling range (*11*–*18*) (Fig. 1B). This question has not been resolved because the biophysical properties of diffusion and clearance have not been measured for any activator/inhibitor pair.

The transforming growth factor–β superfamily signals Nodal and Lefty constitute an activator/inhibitor-based system in animals as different as sea urchin and mouse (*3*–*5*, *16*, *18*–*22*) (text S2). Nodals activate signaling during mesendoderm induction and left-right patterning,

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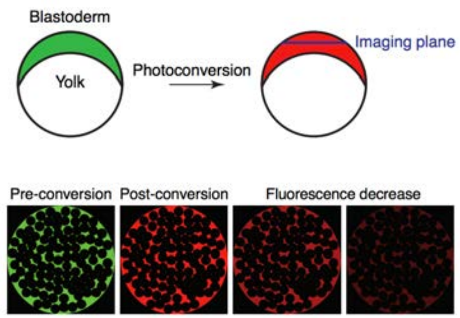
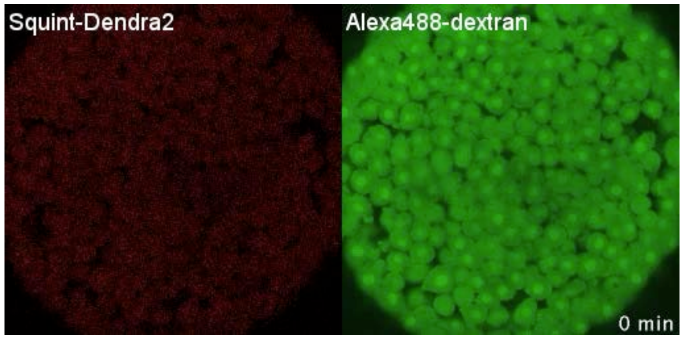
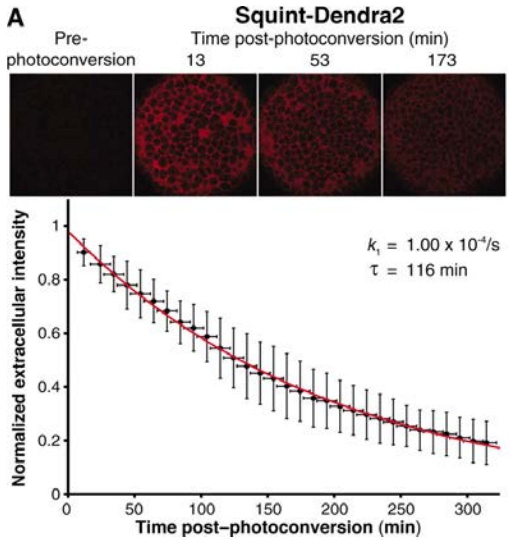
†These authors contributed equally to this work.

Measuring extracellular clearance rate constants with pulse-labelling assay

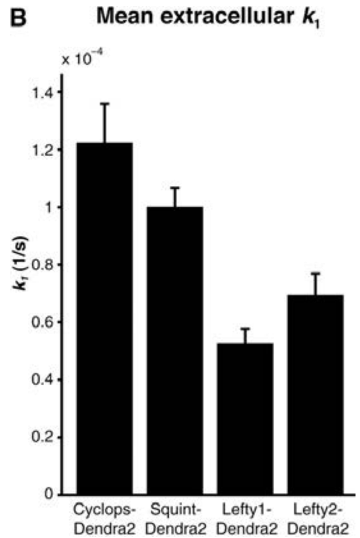




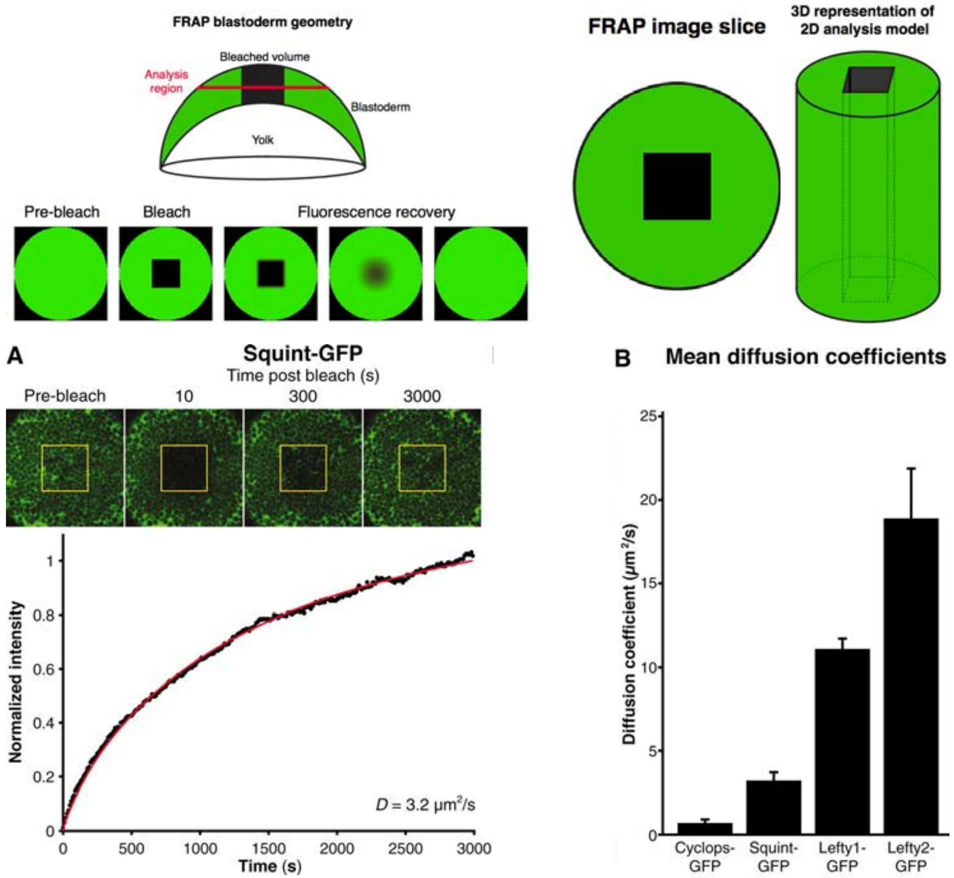


$$\frac{dc}{dt} = -k_1 c$$

$$\bar{\tau} = \frac{\ln(2)}{\bar{k}_1}$$



Measuring effective diffusion coefficients using Fluorescence Recovery After Photobleaching (FRAP)



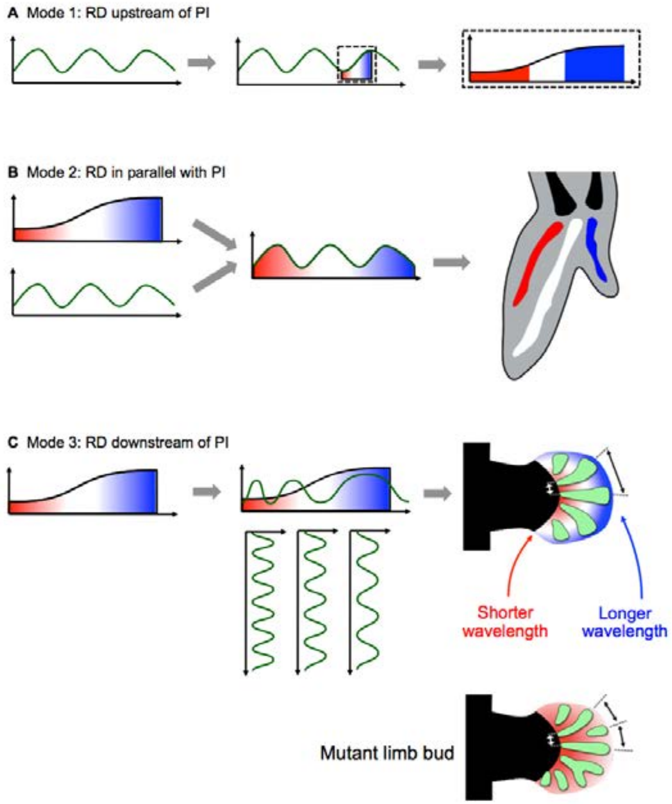
Conclusion

Differential diffusivity underlies differences in activator/ inhibitor range

Similar trends in half-lives, but differences in diffusivity are much more pronounced compared to differences in clearance

Previously described network topology and biophysical properties support **activator/inhibitor reaction-diffusion model of morphogenesis**

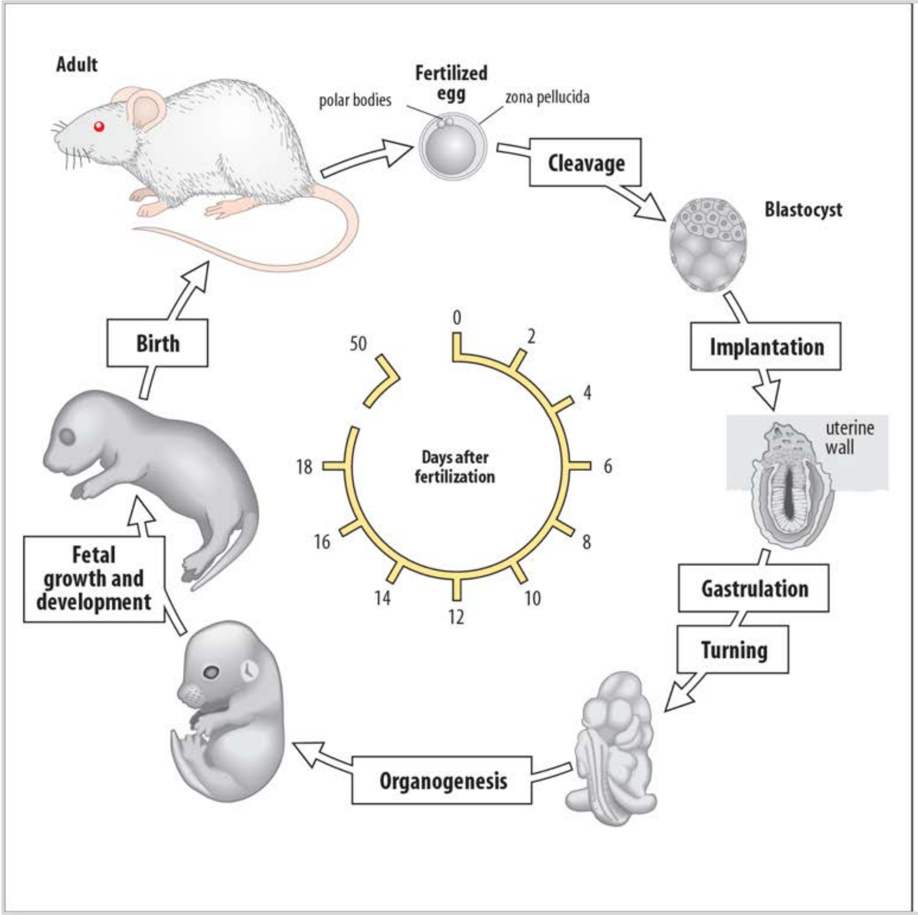
Modes of possible reaction-diffusion (RD)/ positional information (PI) interaction

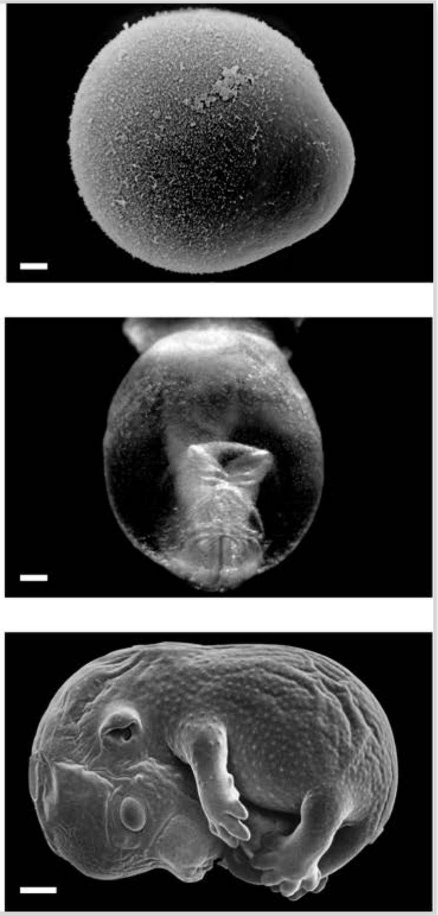


What we will cover

- Quantitative imaging of transcription factor dynamics in mouse (mus musculus) embryos

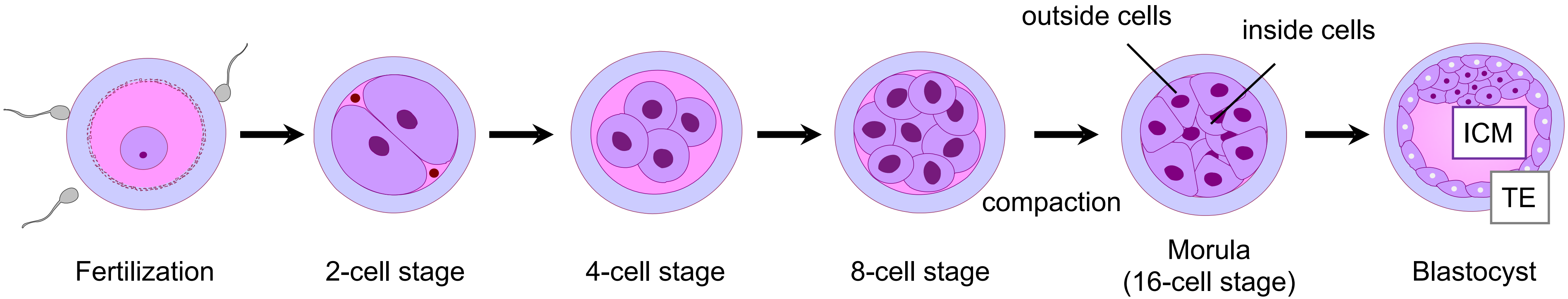
The life cycle of the mouse





How does a single cell develop into a complex multicellular organism?

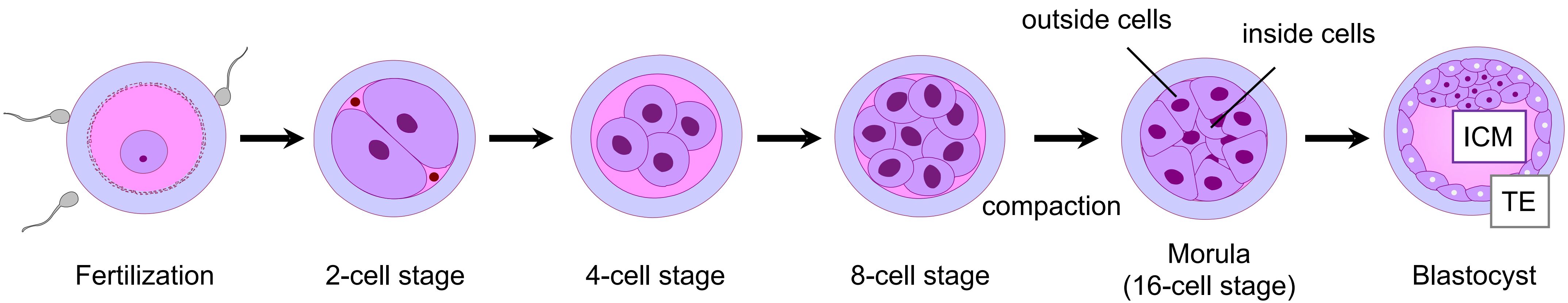
Early mammalian embryo development



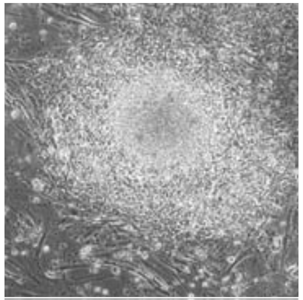
ICM = Inner cell mass (-> Embryo)
TE = Trophectoderm (-> Placenta)

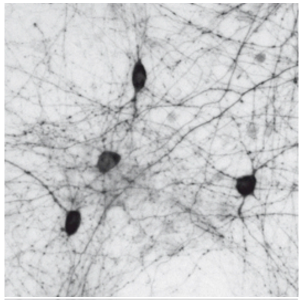
How does a single cell develop into a complex multicellular organism?

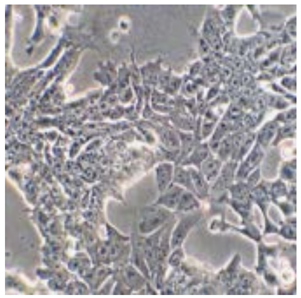
Early mammalian embryo development



Why is this a fundamental question?

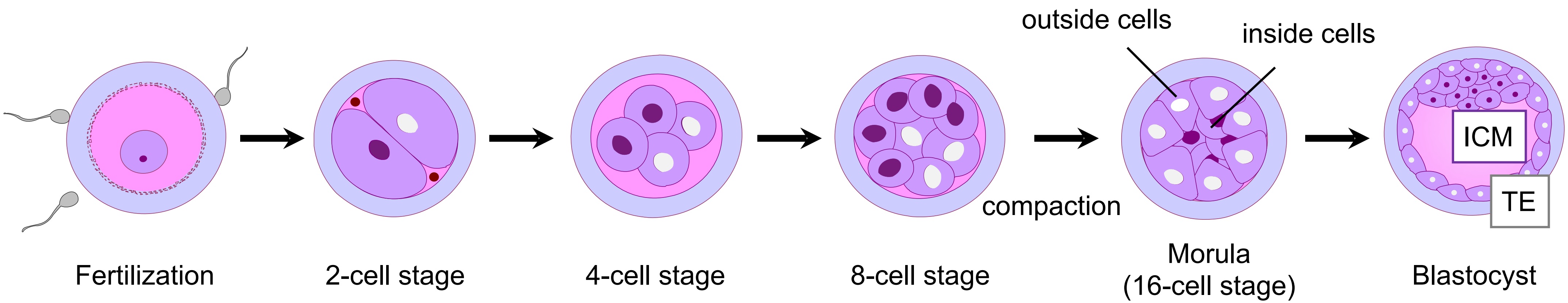

iPS Cells


Somatic Cells


ES Cells

How does a single cell develop into a complex multicellular organism?

Early mammalian embryo development



Models of asymmetry generation

‘Stochastic’ model
First asymmetry at the 16-cell stage.

‘Biased heterogeneity’ model
Bias does exist before compaction.

Only static readouts

When is patterning first detectable in mammalian embryos?

Int. J. Dev. Biol. 50: 581-588 (2006)
doi: 10.1387/ijdb.062181th

Developmental Perspectives

Where do we stand now? – mouse early embryo patterning meeting in Freiburg, Germany (2005)

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Freiburg Meeting Participants (in alphabetical order): Vernadeth Alarcon, Jens-Erik Dietrich, Toshihiko Fujimori, David Glover, Takashi Hiiragi, Nihan Kara, Sophie Louvet-Vallee, Marek Maleszewski, Yusuke Marikawa, Bernard Maro, Nami Motosugi, Hitoshi Niwa, Zbigniew Polanski, Berenika Plusa, Davor Solter, Yayoi Toyooka, Akiko Tsumura, Masamichi Yamamoto, Magdalena Zernicka-Goetz. (Note: Richard Gardner did not join the meeting, but was of course invited to participate.) The invited moderators of the meeting were (in alphabetical order): Chris F. Graham, Jacek Kubiak and Andrzej K. Tarkowski.

Note from the Editor-in-Chief: This *Developmental Perspecitve* paper presents the reflections of its authors on the controversial subject of early embryo patterning in the mouse, on the basis of an international meeting, celebrated in Freiburg, Germany on 16th September 2005. It serves as a record of the meeting (though not strictly speaking, a Meeting Report), detailing the contentious issues that were discussed and highlights the unsettled nature of our current understanding of these issues.

Peer-reviewing of this manuscript elicited contradictory and somewhat passionate responses. It even came to light that a former version of this manuscript had not been considered to be suitable for publishing in a number of other prestigious journals in the field, due to claims of being appalingly imbalanced, misleading and even alarmist.

Mechanism underlying mammalian preimplantation development has long been a subject of controversy and the central question has been if any "determinants" play a key role in a manner comparable to the non-mammalian "model" system. During the last decade, this issue has been revived (Pearson, 2002; Rossant and Tam, 2004) by claims that the axes of the mouse blastocyst are anticipated at the egg ("prepatterning model"; Gardner, 1997; Gardner, 2001; Piotrowska *et al.*, 2001; Piotrowska and Zernicka-Goetz, 2001; Zernicka-Goetz, 2005), suggesting that a mechanism comparable to that operating in non-mammals may be at work. However, recent studies by other laboratories do not support these claims ("regulative model"; Alarcon and Marikawa, 2003; Chroscicka *et al.*, 2004; Hiiragi and Solter, 2004; Alarcon and Marikawa, 2005; Louvet-Vallee *et al.*, 2005; Motosugi *et al.*, 2005) and the issue is currently under hot debate (Vogel, 2005). Deepening our knowledge of this issue will not only provide an essential basis for understanding mammalian development, but also directly apply to ongoing clinical practices such as intracyto-

Int. J. Dev. Biol. 50: 587 (2006)
doi: 10.1387/ijdb.062181dg

Unfair debate

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Int. J. Dev. Biol. 50: 586 (2006)
doi: 10.1387/ijdb.062181jk

Science is not a democracy

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"La parole ne représente parfois qu'une manière, plus adroite que le silence, de se taire"#
Simone de Beauvoir (1908-1986)

Science is not a democracy. Even if a huge majority of the meetin participants did not agree with the results or interpretations of th Zernicka-Goetz lab, the definitive answer as to whether there any hereditary bias in mouse development needs hard dat obtained independently in different laboratories. Before such har data is obtained, we can only discuss the relevant issues, as w did in Freiburg. For that reason it seems premature to state th such a hereditary bias really exists in mouse early embryo, a stated by recent publications (e.g. Lawrence and Levine, 200

When is patterning first detectable in mammalian embryos?

NATURE|Vol 442|13 July 2006 BRIEF COMMUNICATIONS ARISING

EMBRYOLOGY

Does prepatterning occur in the mouse egg?

Arising from : B. Plusa *et al. Nature* **434,** 391–395 (2005)

BRIEF COMMUNICATIONS ARISING NATURE|Vol 442|13 July 2006

464–469 (2005).
10. Motosugi, N., Bauer, T., Polanski, P., Solter, D. & Hiiragi, T. *Genes Dev.* **19,** 1081–1092 (2005).
11. Alarcon, V. B. & Marikawa, Y. *Mol. Reprod. Dev.* **72,** 354–361 (2005).
12. Fujimori, T., Kurotaki, Y., Miyazaki, J. & Nabeshima, Y. *Development* **130,** 5113–5122 (2003).
13. Piotrowska-Nitsche, K. & Zernicka-Goetz, M. *Mech. Dev.* **122,** 487–500 (2005).
14. Squirrell, J. M., Wokosin, D. L., White, J. G. & Bavister, B. D. *Nature Biotechnol.* **17,** 763–767 (1999).
15. Vinot, S. *et al. Dev Biol.* **282,** 307–319 (2005).

doi:10.1038/nature04907

EMBRYOLOGY

Plusa et al. *reply*

Replying to: T. Hiiragi, S. Louvet-Vallée, D. Solter & B. Maro *Nature* **442,** doi:10.1038/nature04907 (2006)

blastocyst parts are consistent with independent findings[4,5,7]. In assessing this, we found that it was critical to count the proportion of cells contributing to different blastocyst parts: to be significant, this had to be more than 70%[13], and not simply three or fewer cells[1].

Most important, if cleavage were random and blastomere fate could not be predicted, as Hiiragi *et al.* suggest, we would not have been able to isolate four-cell-stage blastomeres of like identity from different embryos and combine them to make chimaeras with different developmental properties[8]. Further weaknesses in the case against prepatterning are discussed elsewhere[14].

When is patterning first detectable in mammalian embryos?

This article has been retracted

Science 17 February 2006:
Vol. 311 no. 5763 pp. 992-996
DOI: 10.1126/science.1120925

REPORT

Cdx2 Gene Expression and Trophectoderm Lineage Specification in Mouse Embryos

Kaushik Deb[1], Mayandi Sivaguru[2], Hwan Yul Yong[1] and R. Michael Roberts[1,3,*]

+ Author Affiliations

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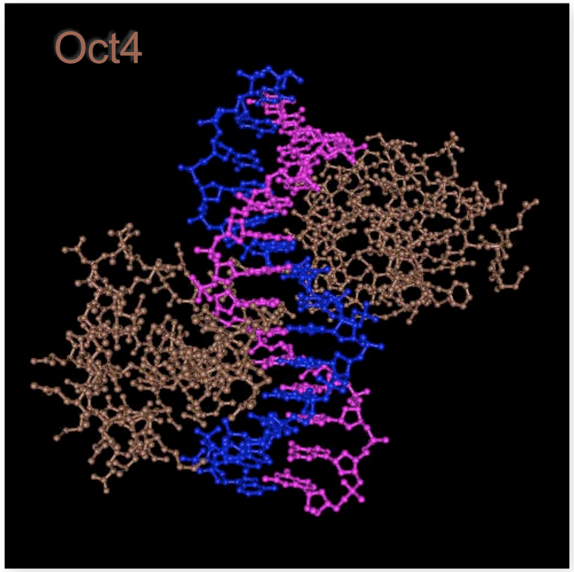
ABSTRACT

Controversy exists as to whether individual blastomeres from two-cell-stage mouse embryos have identical developmental properties and fate. We show that the transcription factor Cdx2 is expressed in the nuclei of cells derived from the late-dividing but not the first-dividing blastomere of two-cell embryos and, by lineage tracing and RNA interference knock-down experiments, that this lagging cell is the precursor of trophectoderm. *Cdx2* mRNA is localized toward the vegetal pole of oocytes, reorients after fertilization, and becomes concentrated in the late-dividing, two-cell-stage blastomere. The asymmetrical distribution of *Cdx2* gene products in the oocyte and embryo defines the lineage to trophectoderm.

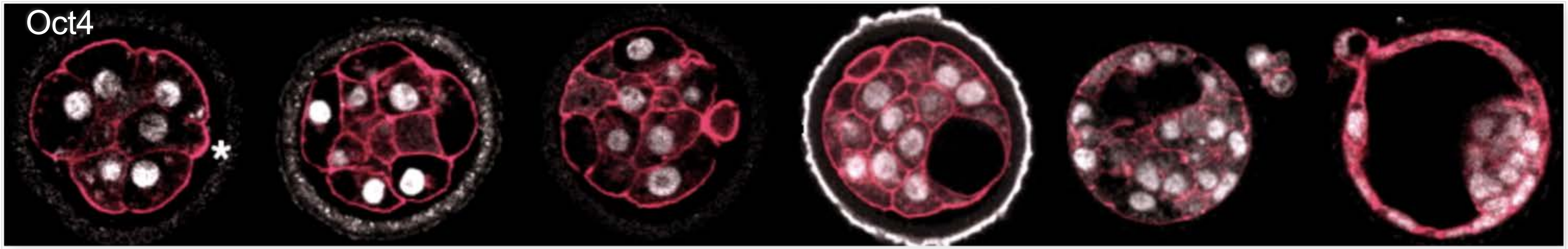
Transcription factor Oct4

- Marker for undifferentiated stem cells
- Pou & HD DNA binding domains
- Expressed in all cells of embryo
- Oct4 mutant affects only ICM cells
- Oct4 levels uncorrelated to ICM/TE lineages

Dynamic Behaviour?

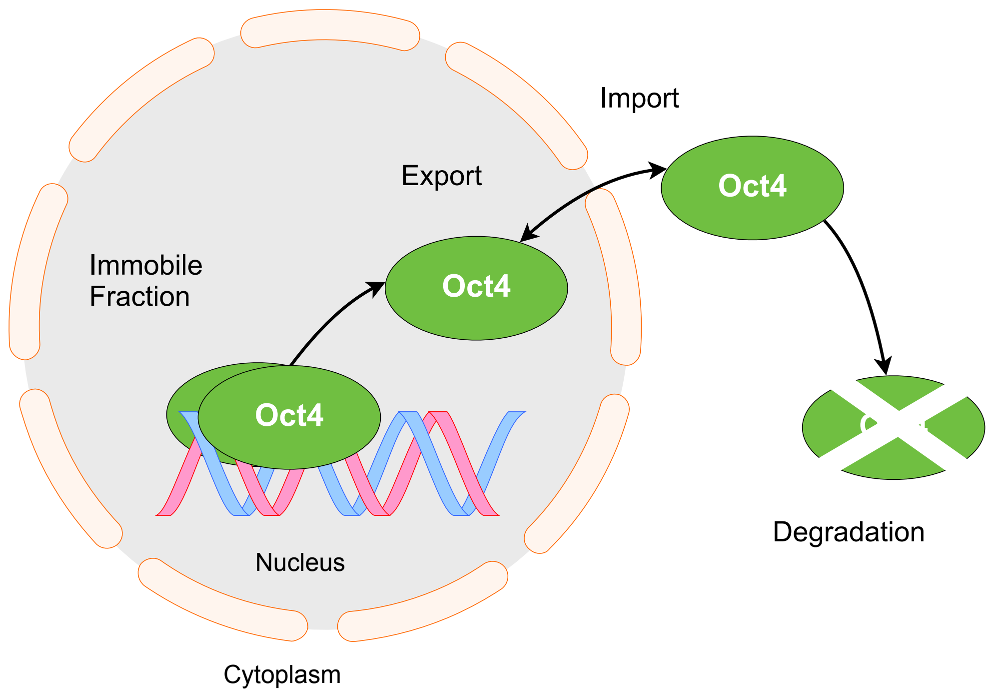


Klemm et al., Cell (1994)

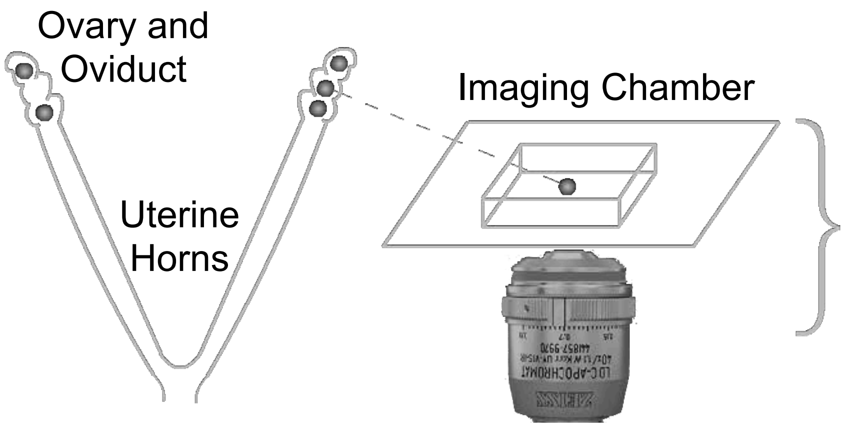


Dietrich&Hiiragi, Development (2007)

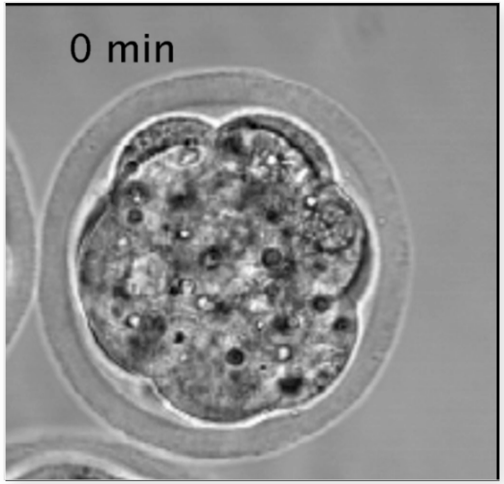
Transcription factor dynamics: Export-Import-Degradation-Immobile fraction



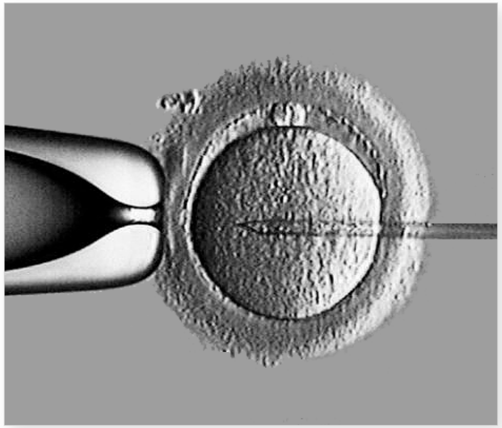
Imaging mouse embryos



Laser Scanning Microscope

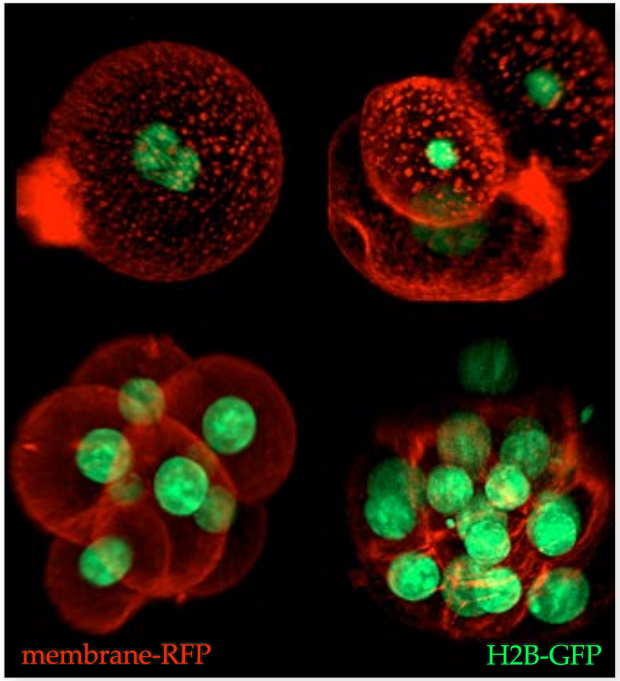


Rapid genetic manipulations in mouse embryos



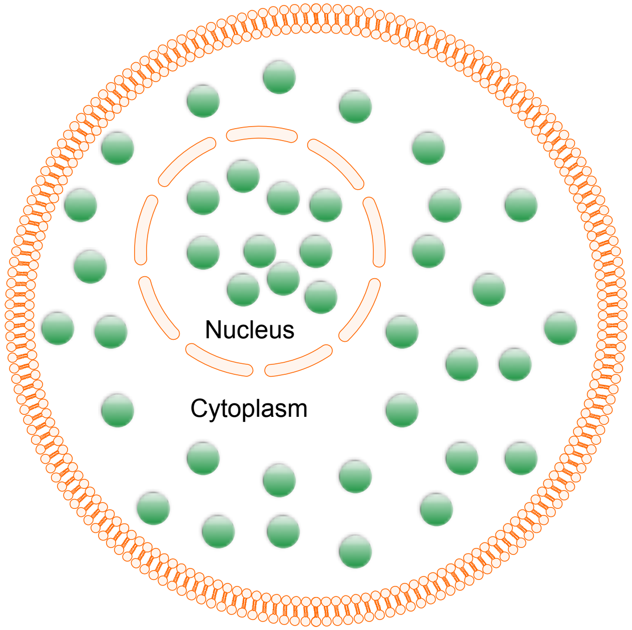
H2B (nuclear) - GFP (Green Fluorescent Protein)

Lck (membrane) - RFP (Red Fluorescent Protein)



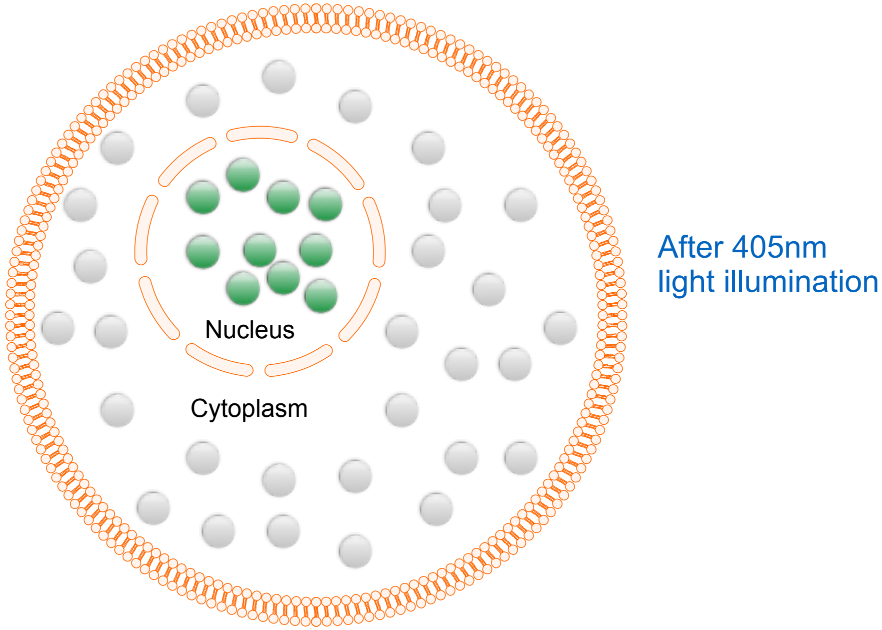
Imaging protein dynamics in single cells of the embryo: global versus selective fraction

GFP



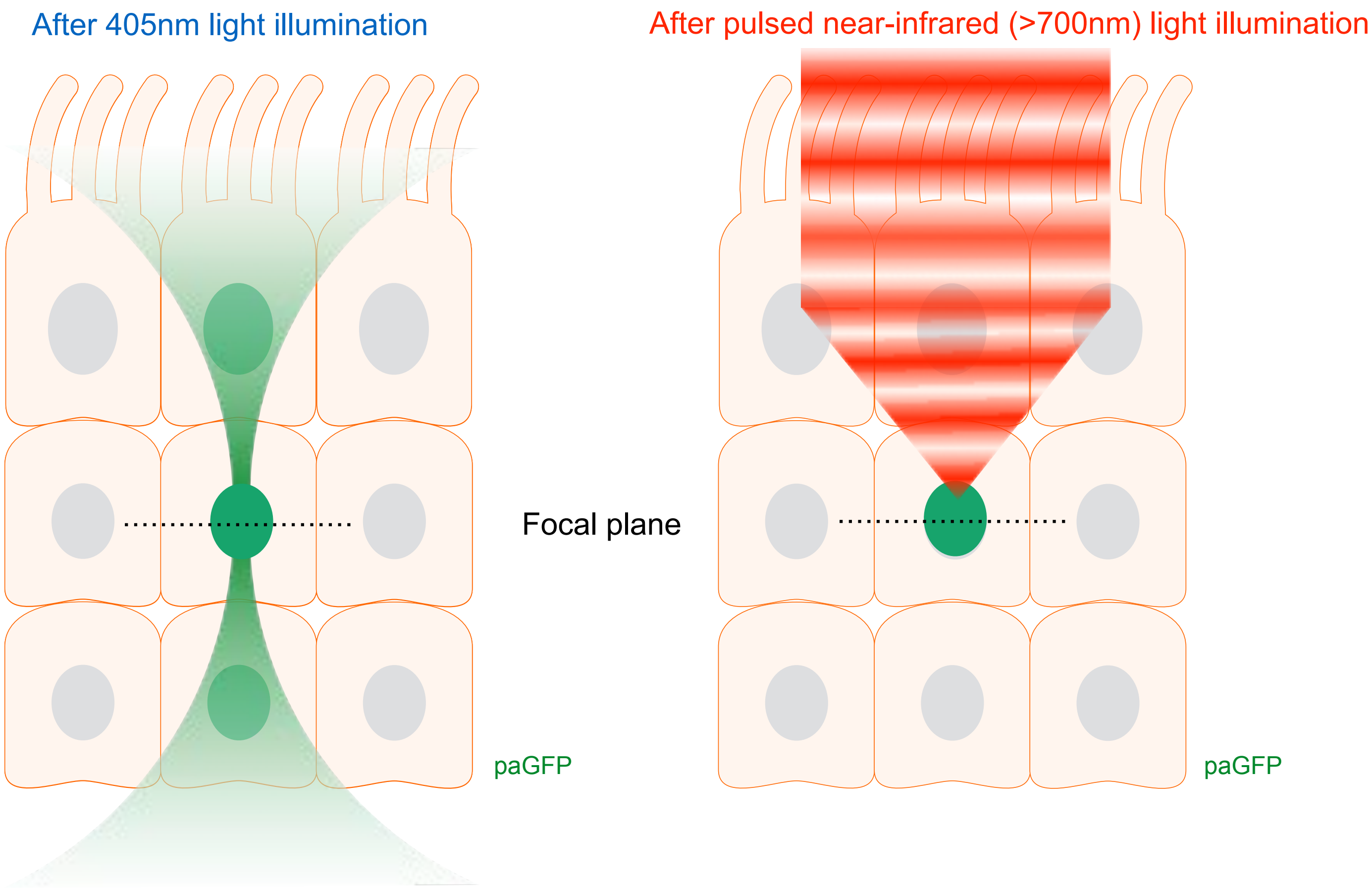
Entire protein population labeled

paGFP (photoactivatable GFP)

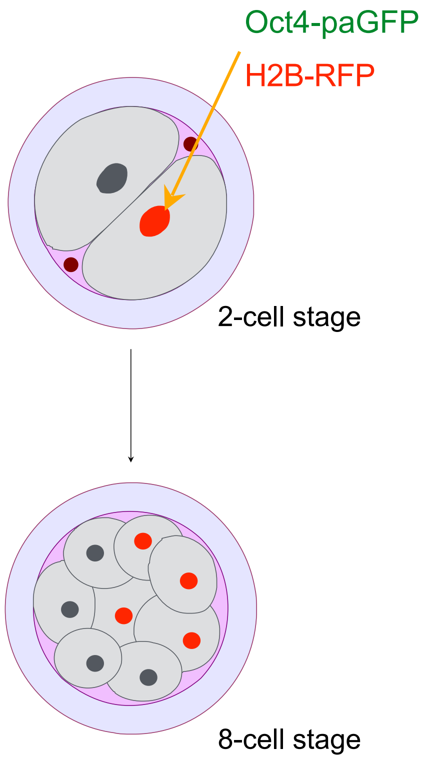


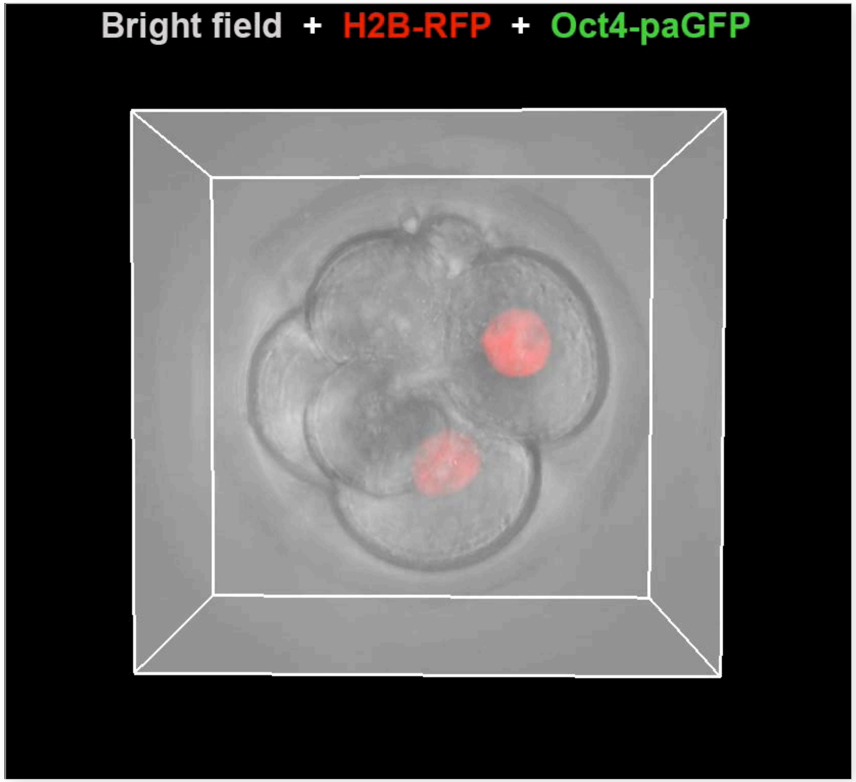
Selective protein population labeled

Achieving selective photoactivation in 3D

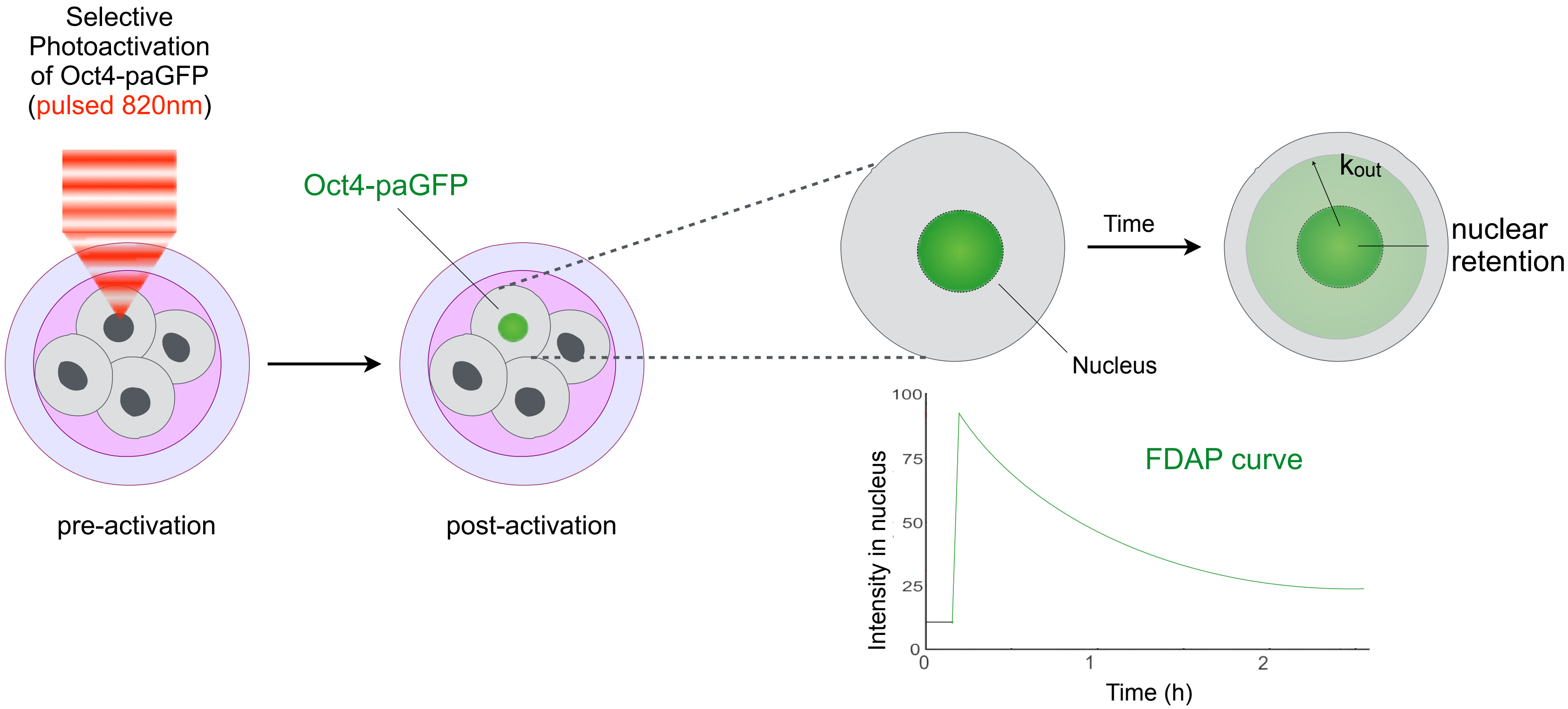


Oct4-paGFP photoactivation in single cells of the mouse embryo

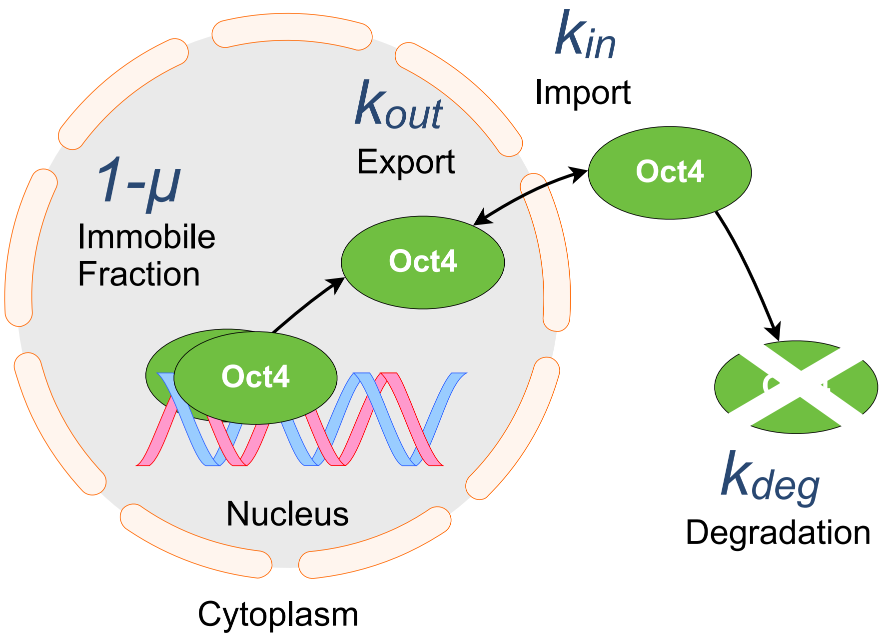




Fluorescence Decay After Photoactivation (FDAP) of photoactivatable Oct4



Transcription factor dynamics:
Export-Import-Degradation-Immobile fraction



Description of Oct4-paGFP dynamics

$$\frac{d}{dt}Oct4_n = k_{in}Oct4_c - \left(k_{out} + k_{deg}\right)Oct4_n$$

Export rate k_{out} & immobile fraction $1\text{-}\mu$

$$f(t) = \mu f_0 e^{-\left(k_{out}+k_{deg}\right)t} + (1-\mu) f_0$$

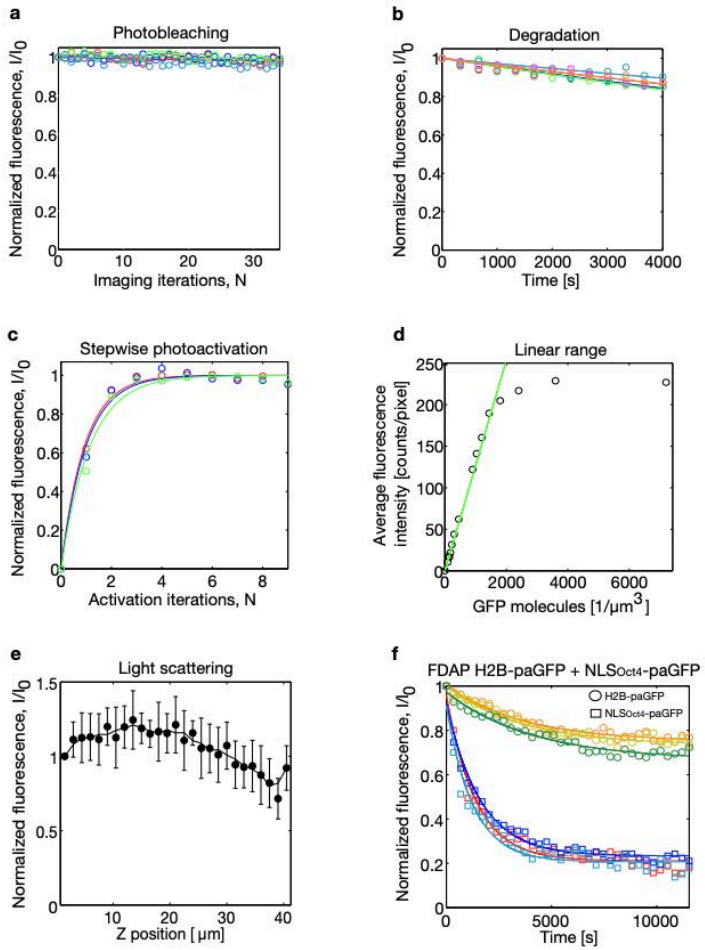
Degradation k_{deg}

$$f(t) = f_0 e^{-k_{deg}t}$$

Import rate k_{in}

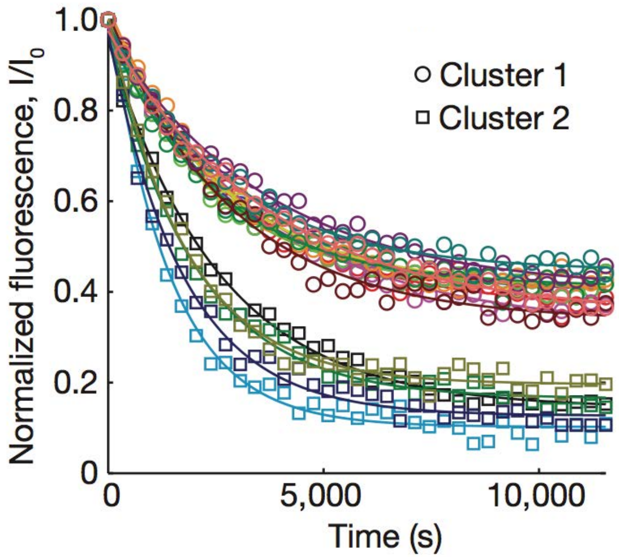
$$k_{in} = \left(k_{out} + k_{deg}\right)\mu \frac{Oct4_n^{total}}{Oct4_c}$$

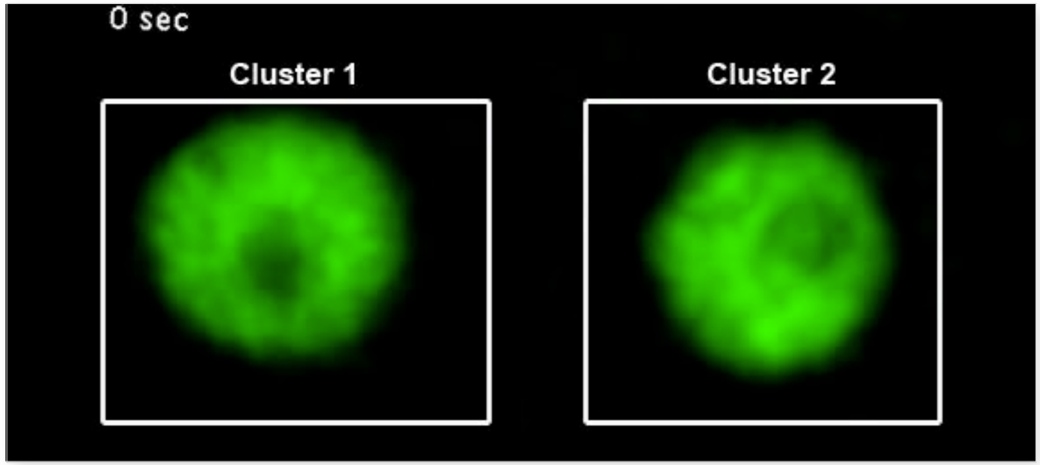
FDAP Assay: Imaging controls



Cells fall into two clusters of distinct kinetic behaviours prior to compaction

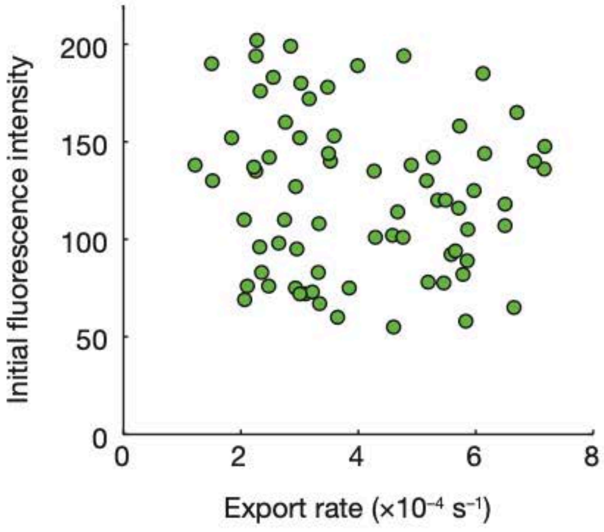
FDAP curve for
4- and 8-cell stage embryos

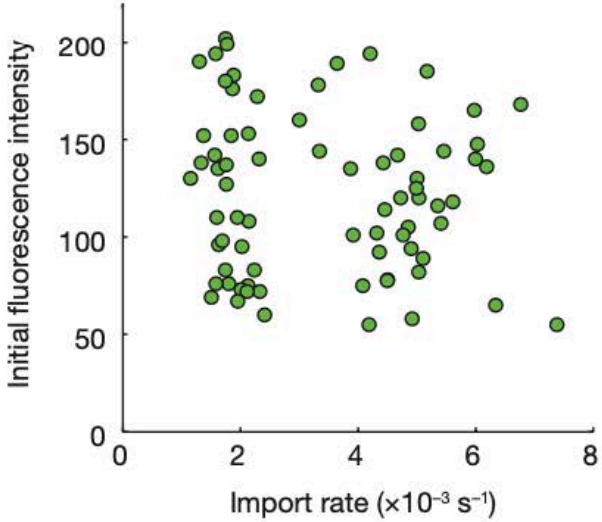


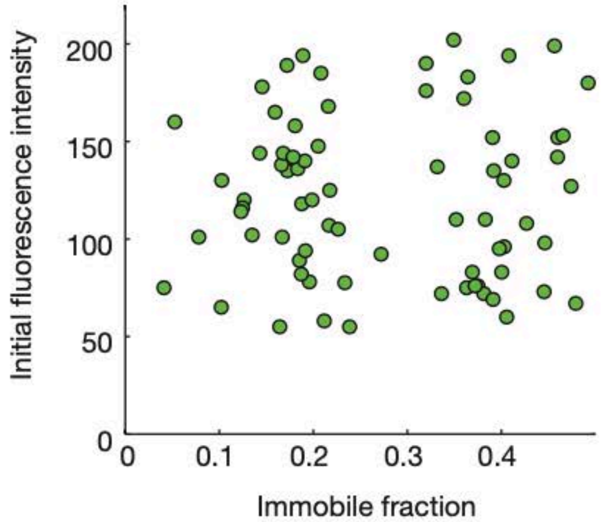


Oct4-paGFP

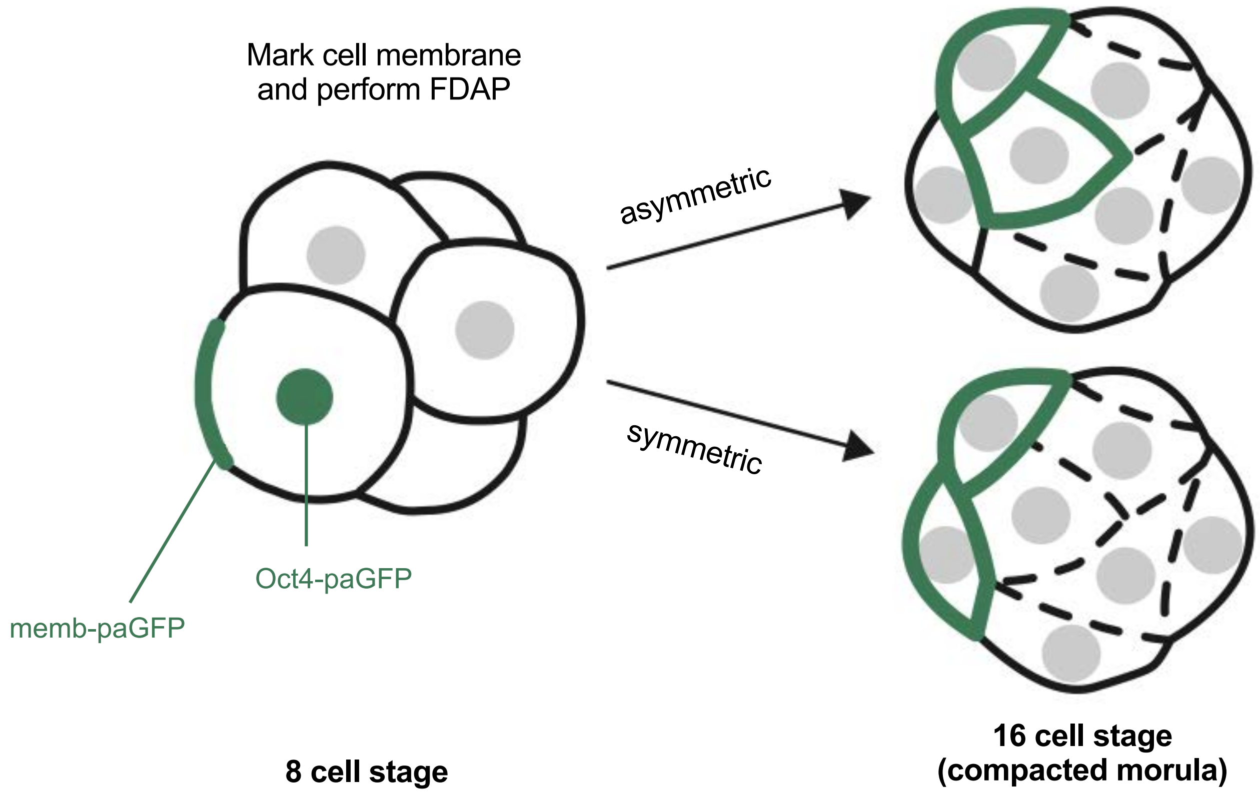
Kinetics are independent of protein expression levels



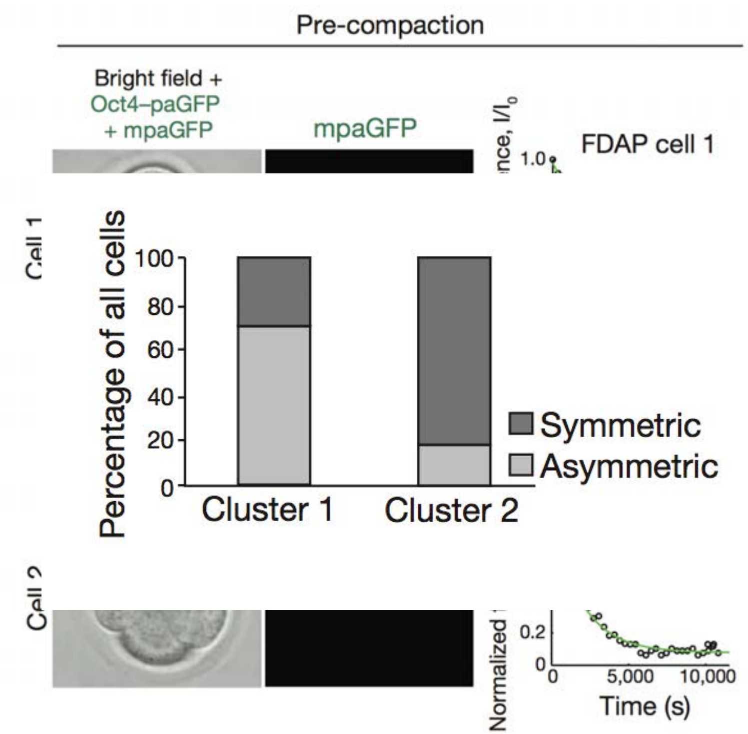


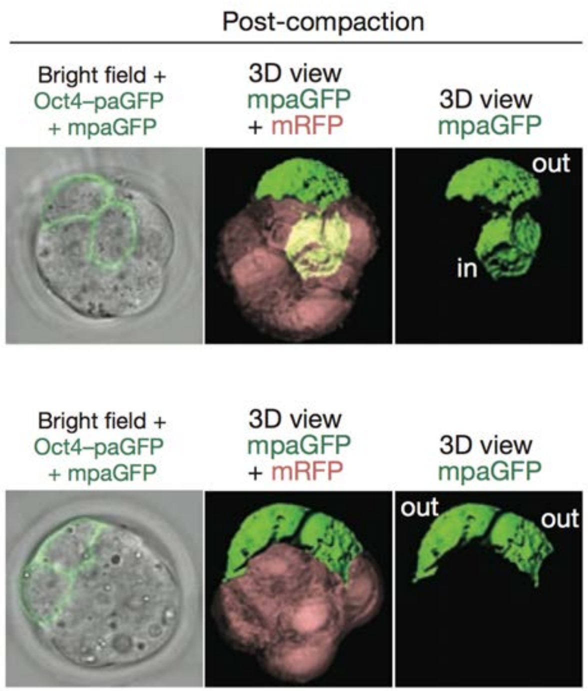


Do distinct Oct4 kinetic behaviours predict embryonic patterning events?

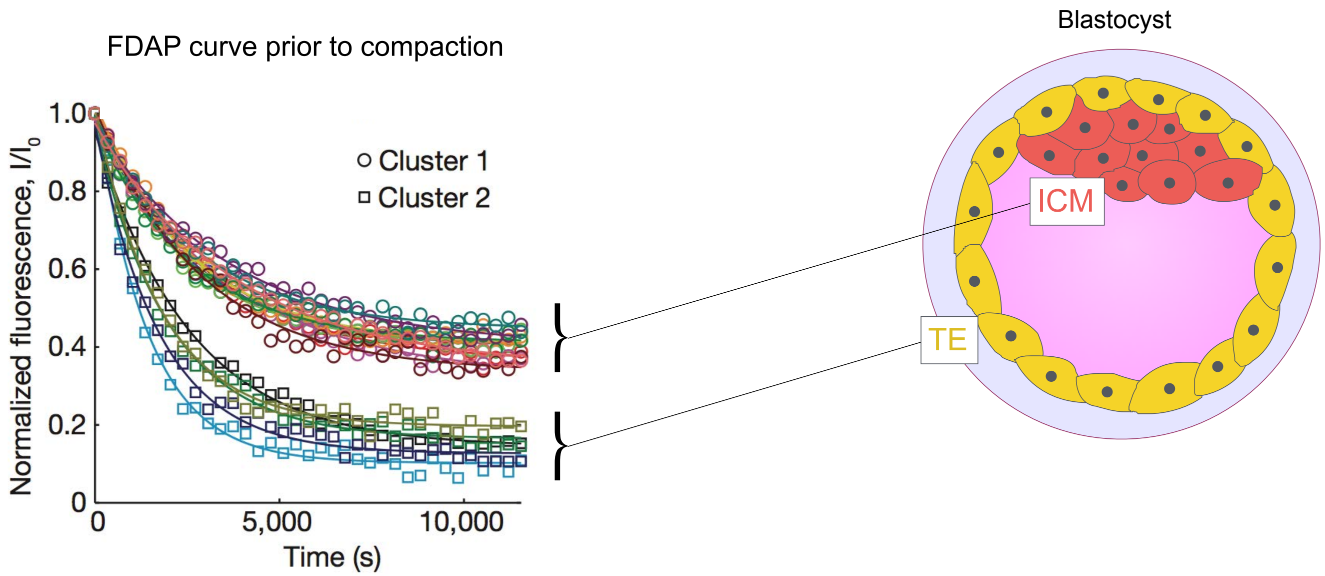


Oct4 kinetics predict whether cells divide asymmetrically or symmetrically



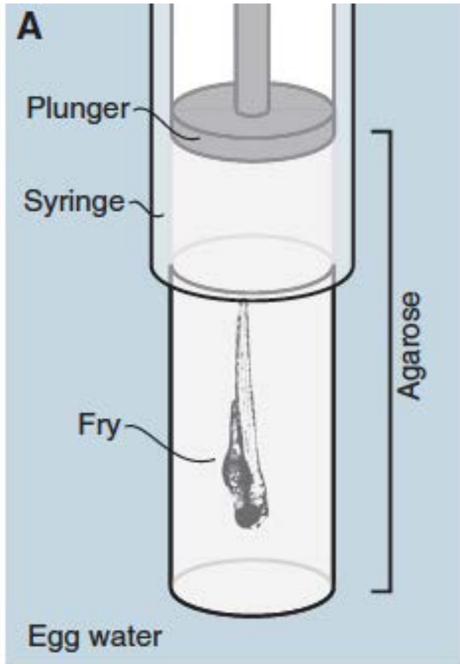


Oct4 kinetics predict ICM/TE lineage patterning in the early mammalian embryo



Conventional light-sheet setup

- Conventional
- Specimen vertical
- Embedded in Agarose or capillaries
- No mammalian embryonic development
- Parallel imaging of multiple embryos difficult



Inverted light-sheet microscope for imaging mouse pre-implantation development

Petr Strnad[1], Stefan Gunther[1], Judith Reichmann[1], Uros Krzic[1], Balint Balazs[1], Gustavo de Medeiros[1], Nils Norlin[1], Takashi Hiiragi[2], Lars Hufnagel[1] & Jan Ellenberg[1]

Despite its importance for understanding human infertility and congenital diseases, early mammalian development has remained inaccessible to *in toto* imaging. We developed an inverted light-sheet microscope that enabled us to image mouse embryos from zygote to blastocyst, computationally track all cells and reconstruct a complete lineage tree of mouse pre-implantation development. We used this unique data set to show that the first cell fate specification occurs at the 16-cell stage.

In toto imaging of biological model organisms such as worm[1], fly[2] and fish[3] has revolutionized the understanding of cell division and cell fate specification in their early development. However, because of high light sensitivity and demanding *in vitro* culture requirements, *in toto* imaging of mammalian pre-implantation embryos has not been possible, and a number of important questions, including the time of cell fate specification, remain controversial as a result[4–12]. In addition, because of the high degree of variability in early mammalian development[13], a statistical analysis of accurately reconstructed cell lineages of several embryos is a prerequisite to drawing solid conclusions. Light-sheet microscopy[14] is the tool of choice to image large and light-sensitive specimens. Because the excitation light is restricted to the focal plane, light exposure and phototoxicity are minimized. For a typical mammalian pre-implantation embryo of 80–100 μm in diameter, this allows the light dose to be reduced by about two orders of magnitude, as compared to standard confocal or two-photon microscopy, without compromising resolution[15]. The need to place the specimen in the narrow space between the illumination and detection objectives, however, has led to light-sheet microscope designs in which the specimen is held vertically by embedding into agarose cylinders[14] or small capillaries[16], which is incompatible with mammalian embryo development. Alternative light-sheet microscope configurations that eliminate the requirement for sample embedding have been developed[1,17–20], but none of them is well suited for high-resolution imaging of many pre-implantation embryos in parallel (**Supplementary Discussion**). Although live imaging of mouse pre-implantation development has been conducted previously by confocal microscopy[21–24], none of these studies successfully imaged complete development from zygote to blastocyst. Furthermore, light-sheet microscopy has so far only been used to image post-implantation mouse embryos[19,20].

To overcome this limitation, we developed an inverted light-sheet microscope in which both the illumination and the detection objective face upward in an immersion water reservoir (**Fig. 1a** and **Supplementary Fig. 1**). The specimen is held by gravity in a long, narrow channel of a microscope slide–sized holder (**Fig. 1b**) and can therefore be lowered between the two objectives (**Fig. 1c**), much like a standard glass-bottom dish on an inverted microscope. Because the transparent plastic of the sample channel has the refractive index of water, the specimen is optically fully accessible for high-resolution immersion objectives, yet it is physically isolated from the immersion medium and can be maintained in standard microdrop *in vitro* embryo culture (**Fig. 1c**). Full environmental incubation of the imaging system (**Supplementary Fig. 2**) allows maintenance of stable temperature and atmospheric composition. Furthermore, free movement along the long axis of the specimen channel makes it possible to place many embryos in a row for high-throughput imaging (**Fig. 1b**). For maximum multicolor imaging speed, we implemented simultaneous imaging of separated EGFP and mCherry fluorescence signals on a single camera chip (**Supplementary Fig. 3**). Our asymmetric optical design, with a lower-numerical-aperture light sheet–generating objective and a high-collection-efficiency detection objective, provides resolution comparable to that of confocal microscopy (**Supplementary Fig. 4**) at much lower light dose.

Mammalian development from the fertilized zygote to the fully grown blastocyst with 64 cells requires approximately 3 d. At this time, the first cell fate decisions have been taken and the first two early embryonic lineages are segregated: the trophectoderm (TE), which forms extraembryonic tissue, and the inner cell mass (ICM), which gives rise to the embryo proper and is the source of embryonic stem cells[25]. The two lineages can be clearly distinguished in the blastocyst as only TE cells are in direct contact to the outside environment. For *in toto* reconstruction of pre-implantation development, it is necessary to track all cells through the first six divisions, for which we chose the chromosomal protein histone 2B (H2B) tagged with a red fluorescent protein (mCherry) as a marker[26]. To assign cell fates at the blastocyst stage, we crossed mice with the chromosomal marker

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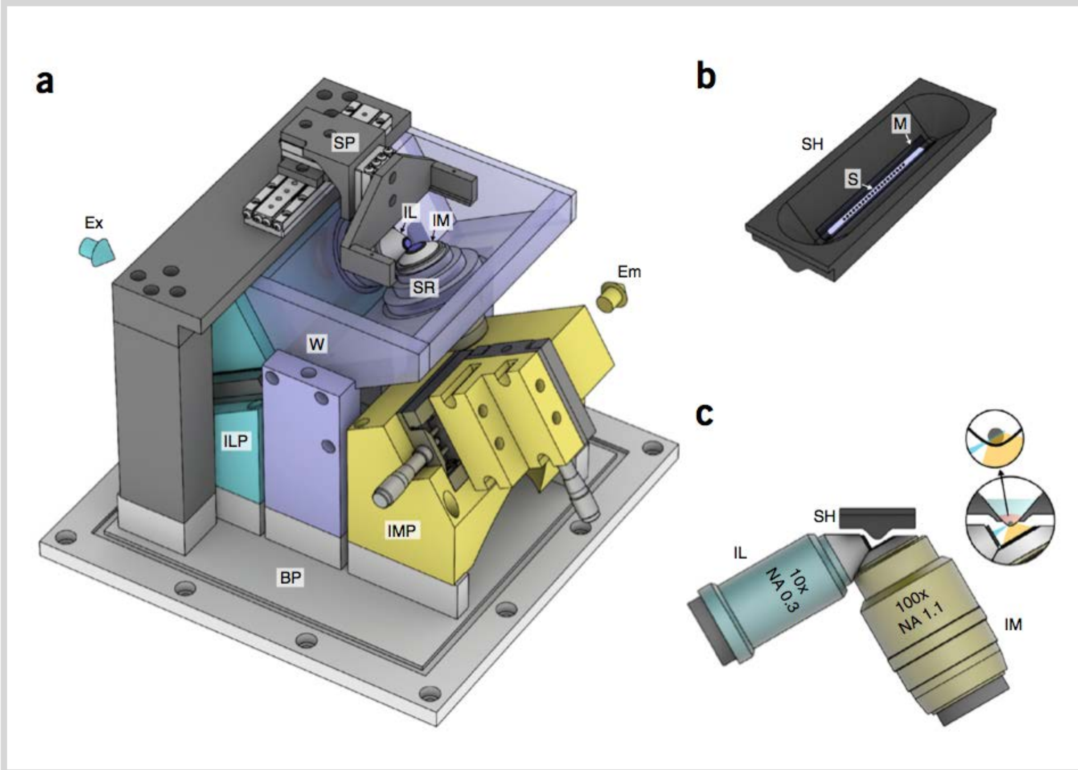
Inverted light-sheet microscope

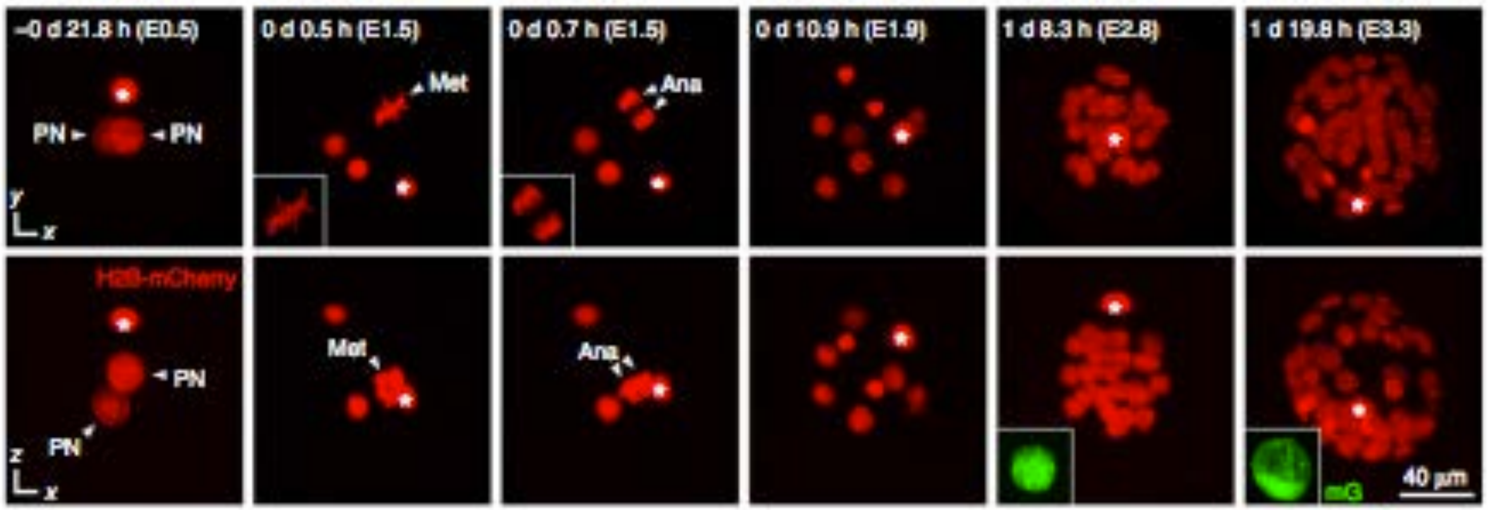
- Inverted
- Specimen horizontal
- Submerged in water
- Mammalian embryonic development possible
- Many embryos in parallel



Volumetric imaging of mice embryos

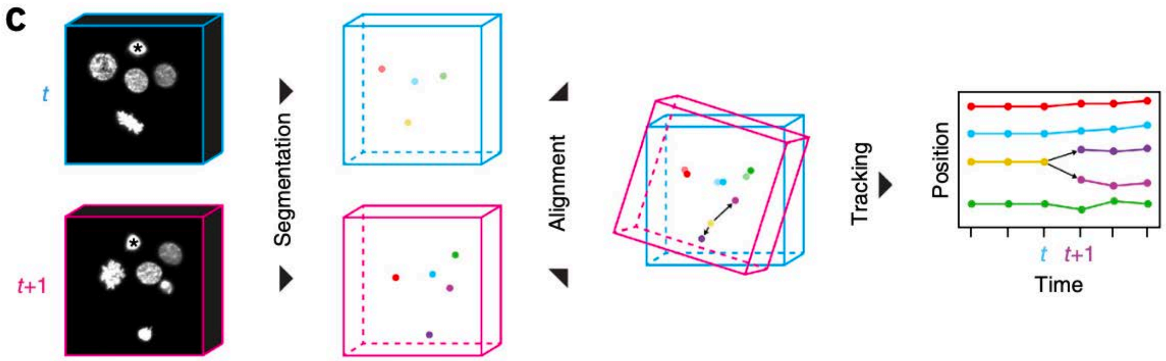
- Crossed mice with a chromosomal protein histone B2 (H2B) tagged mCherry to mice expressing a cell surface marker (GFP with myristylation sequence)

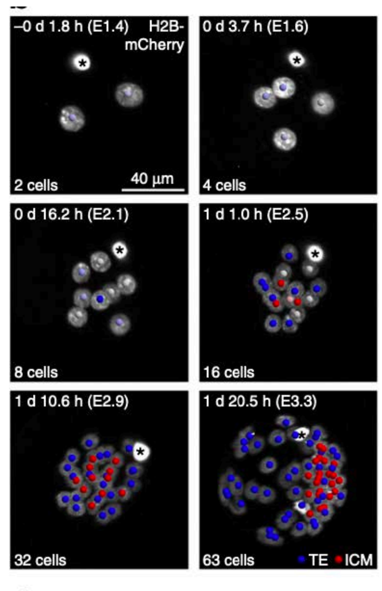
- Imaged 139 embryos
  - Only 12 were double heterozygotes and had fully analysable data sets
  - Excluded rapidly rotating embryos and embryos that moved out of the field of view



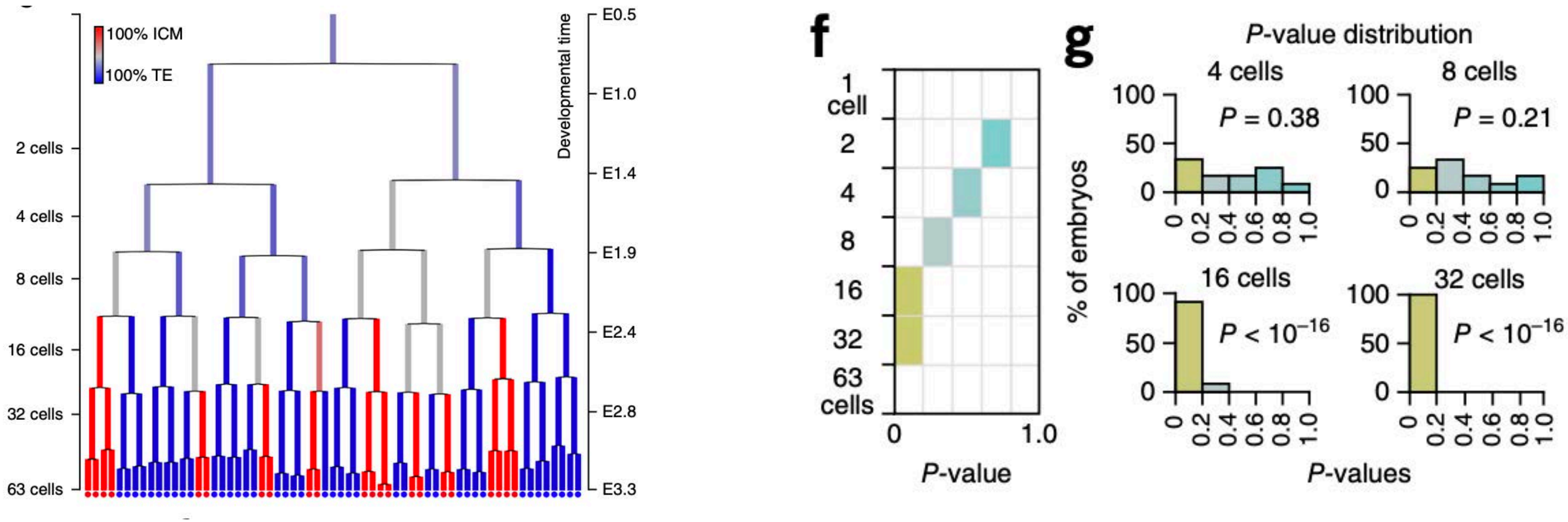
Derive cell trajectories from data

- Two step segmentation and tracking pipeline
- Backtracked cell lineages





Cell fate is biased toward ICM or TE lineage from the 16-cell stage



- 16-cell embryo is the earliest stage at which blastomeres are significantly biased in their contributions to TE or ICM cell fate

Imperial College London

05 December 2024

Thank you!

Periklis (Laki) Pantazis